Default Depressions*

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Abstract

Prolonged sovereign defaults are followed by economic depressions that persist well after their resolution. Short defaults do not have this scarring effect. I develop a model explaining this asymmetry alongside standard sovereign risk dynamics. It incorporates complementarities in technology adoption from the "big push" literature in a unique-equilibrium setting. Defaults disrupt adoption and act as a "big pull" force that shifts the economy from a "rich" to a "poor" state. These dynamics generate a bimodal ergodic distribution with fluctuations between "normal times" and a "default trap", characterized by lower capital and technology adoption, more frequent defaults, and higher spreads.

Keywords: Sovereign default, Economic depressions, Big push.

JEL Codes: F34, F41, O11, E61

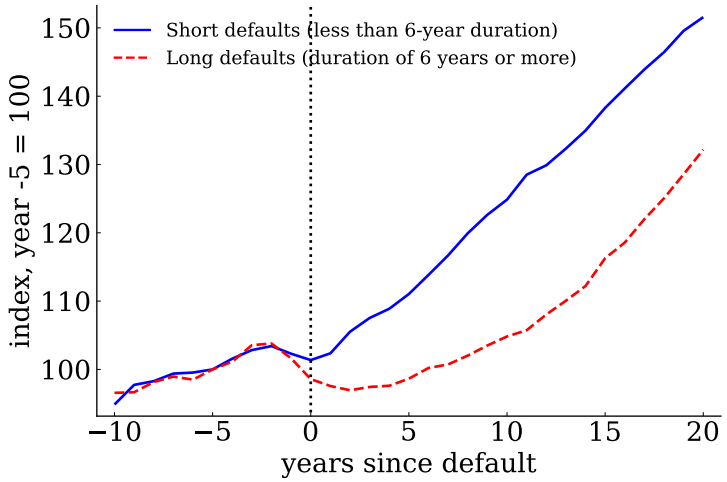
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1 Introduction

Sovereign defaults are large and disruptive events, but some are more disruptive than others. Long-lasting defaults have scarring effects on output and development that are not observed after shorter ones. Figure 1 illustrates this by sorting 213 default episodes into two groups: long defaults with a duration of 6 years or more (the median), and short defaults with a duration of less than 6 years. The data are the average paths of real GDP per capita, with period 0 denoting the year of default.

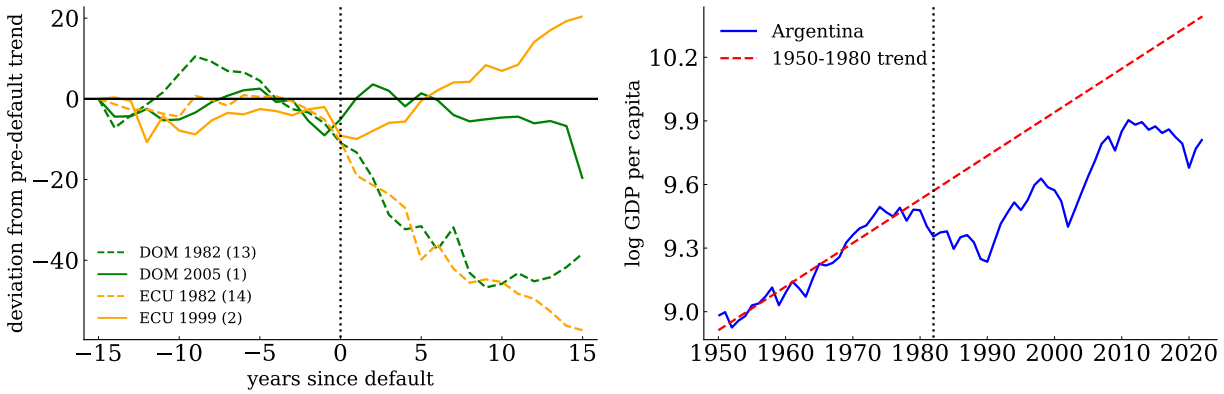
Figure 1: Average real GDP per capita around short and long defaults



Note: Average real GDP per capita, normalized so that the year 5 before each default event is equal to 100. Real GDP per capita is from the Maddison Project Database (Bolt and van Zanden, 2024), and default events and their durations are from Reinhart and Rogoff (2009). The data include 213 default events between 1807 and 2013.

Both groups experience a contraction in GDP around the default event. The recoveries, however, are significantly different. Output quickly bounces back to the pre-default trend in the case of short defaults, while it remains depressed following long ones. One could argue that countries that experience long defaults are also countries with chronically lower productivity growth rates, which would explain this long-run divergence, since the two groups consist of different countries. The left panel of Figure 2 shows that this is not necessarily the case by comparing two pairs of long and short defaults: 1982 and 2005 in the Dominican Republic, and 1982 and 1999 in Ecuador. The labels include the year of default and, in parentheses, its duration in years. In these four cases, it is clear that output remains around its pre-default trend following short defaults but significantly diverges after long ones.

Figure 2: Real GDP per capita relative to its pre-default trend



Note: The left panel shows 100 times the log deviations of real GDP per capita from its pre-default linear trend. The labels denote the country, the year of the default, and its duration in parentheses. The right panel shows the natural logarithm of Argentine real GDP per capita and its pre-default linear trend.

The right panel of Figure 2 shows the evolution of GDP per capita in Argentina around its 1982 default. This crisis lasted for 12 years and is one of the great depressions of the twentieth century studied in Kehoe and Prescott (2007). In their chapter, Kydland and Zarazaga (2007) show that the neoclassical growth model accounts well for this lost-decade depression but not for the slow capital accumulation during the expansion that followed. In particular, they note that the evolution of capital remained well below the "bounce-back" recovery that the model would predict, and conjecture that this is because Argentina switched to a regime of high capital taxes in the form of recurring defaults and other forms of confiscation. In favor of this regime-switching conjecture is the fact that Argentina also defaulted on its debt in 1951 and 1956, but these were short-lived events and their effect on output is barely noticeable. In contrast, the default events after 1982—one in 2001 and another in 2014—are associated with significantly deeper contractions. In Section 2, I present further evidence that longer defaults are associated with deeper contractions and that, if they are long enough, they have long-lasting effects that suggest a form of regime switch like the one conjectured by Kydland and Zarazaga (2007) for Argentina. I also present evidence of the effect of default on technology adoption, which is at the core of the mechanism in the model that I develop in Section 3.

As is the case with the neoclassical growth model used by Kydland and Zarazaga (2007), in which total factor productivity (TFP) is exogenous, existing models of sovereign default (see Arellano, 2008 and the subsequent literature) are not suited to studying these scarring effects of long

defaults, because they focus on default risk over the business cycle, with output fluctuating around a stable long-run mean. This is also true for richer models with endogenous production (see Mendoza and Yue, 2012) and capital accumulation (see Park, 2017; Gordon and Guerron-Quintana, 2018; Arellano, Bai, and Mihalache, 2018) in which business cycles are driven by stationary productivity shocks. Default events in these models may have large effects on output, and capital accumulation may make these effects persistent, but once a default is resolved, the economy eventually reverts to its long-run mean. An important exception is earlier work by Aguiar and Gopinath (2006), who study a model with permanent shocks to trend growth.

The main contribution of this paper is to develop a sovereign default model that features an endogenous switching mechanism between a low-default, high-income regime and a frequent-default, low-income one. At the core of the model is a mechanism through which TFP may remain depressed below its pre-default mean, even years after the resolution of the default. This TFP-driven theory of default depressions is consistent with the findings of Kehoe and Prescott (2007), who show that long-run changes in TFP are crucial for explaining economic depressions. Importantly, the likelihood of a default depression occurring in the model is increasing in the length of the default event, as suggested by the data.

To generate this mechanism, I incorporate elements from the big-push literature into a standard sovereign default model with production. This literature analyzes how, in the context of imperfect competition and aggregate demand spillovers, a big push into industrialization can move an economy from a "bad" (poor) to a "good" (rich) equilibrium. My model incorporates these ideas but features a unique dynamic equilibrium with multiple steady states, similar to the work by Schaal and Taschereau-Dumouchel (2024). In the ergodic distribution, the economy fluctuates around—and occasionally switches between—a regime with a high degree of industrialization and another with a low degree. Sovereign defaults trigger a temporary productivity decline, and long-lasting ones act as a "big pull" force that can pull the economy down from a rich, highly industrialized state to a poor, less industrialized one. Short defaults are less likely to have this scarring effect. On the one hand, this potential big pull endogenously increases the cost of defaulting when the degree of industrialization is high, allowing the economy to sustain high levels of debt with minimal default risk. On the other hand, the cost of defaulting is relatively low for the economy in the poor region with low industrialization. Since more frequent default events in this region reduce the likelihood

of re-industrializing, I refer to this region in the ergodic distribution as a default trap.

The model features a small open economy in which a sovereign government makes borrowing, investment, and default decisions without commitment. The government is more impatient than its constituent households, which reflects political economy frictions and biases its decisions toward debt accumulation to front-load consumption. This is the standard sovereign-default block of the model. The novelty lies in the production side, which features a continuum of monopolistically competitive firms whose productivity is heterogeneous in two dimensions. One dimension is permanent and firm-specific. The other is the result of firms' choices to operate a traditional technology or to pay a fixed cost to access a modern, more productive one. When capital is scarce, only the most productive firms adopt the modern technology. As capital becomes more abundant, production costs decrease and more firms find it profitable to adopt the modern technology. These assumptions generate increasing returns to scale for certain levels of capital and demand complementarities in the sense that, for some firms, the decision to adopt the modern technology depends on how many other firms also do so. As in Buera, Hopenhayn, Shin, and Trachter (2021), ex-ante firm heterogeneity generates a big-push region, where a small increase in the capital stock has a large effect on technology adoption (in their model the main focus is on firm-level distortions and the effects of reducing them through structural reforms). There are other, more stable regions, with significantly lower or higher levels of capital, in which changes to the aggregate capital stock have only moderate effects on adoption. This duality is crucial for the model to generate the bimodal ergodic distribution described above. Another advantage of the ex-ante heterogeneity of firms is that it guarantees a unique production equilibrium under certain conditions on the parameters, which simplifies the analysis by ruling out production multiplicity due to coordination failures.

The only source of risk in the model is a shock to aggregate productivity, which is exogenously penalized during periods of default. While in default, the government is in financial autarky and regains access to financial markets with a constant probability. This exogenous default resolution is standard in the literature and, more importantly, it allows me to perform experiments of different default durations without them interacting with other endogenous variables. The reasons why some defaults last longer than others and why economies face default penalties on productivity are interesting open questions in the literature that are beyond the scope of this paper (for a more detailed treatment of these questions, see Benjamin and Wright, 2009; Bocola, 2016; Dvorkin,

Sanchez, Sapriza, and Yurdagul, 2021; and Arellano, Bai, and Bocola, 2017). While the frictions in the economy could interact with the drivers of default duration in a richer environment—such as one with debt renegotiation—the main mechanisms introduced in this paper would still be present. In the interest of clarity, I abstract from such interactions and simplify the modeling of default duration.

I calibrate the model to Argentina using the subsample of the ergodic distribution in the default trap to target the observed data. Default events in the model have significantly different long-lasting effects on GDP and production decisions. After short defaults, GDP and technology adoption quickly bounce back and GDP remains close to its pre-default level in the long run. After a long default, GDP also bounces back, but to a lower level than before the event, and it remains significantly below its pre-default level several years after the default has been resolved. These different effects of default events generate a bimodal ergodic distribution, which shows that the economy fluctuates around two regimes: normal times and the default trap described above. In normal times the capital stock is relatively high, the fraction of firms that operate the modern technology is close to one, default risk is low, and sovereign spreads are low and feature low variance. When the economy is in the default trap, both the capital stock and technology adoption are significantly lower. In addition, defaults are more frequent and spreads are higher and more volatile. Under the benchmark calibration, the median of the fraction of firms adopting the modern technology in the ergodic distribution is 0.17 and there is full adoption only 34 percent of the time. This indicates that the default trap is more common, which illustrates the absence of a big-push force to counter the big pull of lengthy defaults.

Unlike in most of the big-push literature, there is little scope for the government to intervene and improve economic outcomes in this model. This is because of the big pull that results from the government's relative impatience and its inability to commit to future policy. This, along with the assumption that the government can choose all aggregate allocations directly, makes any analysis regarding domestic policy redundant. Instead, I use the model to analyze the effects of foreign aid implemented after a ten-year default. I evaluate the welfare and re-industrialization implications of three types of aid: an unconditional transfer that can be freely allocated by the government, debt relief, and a capital injection. I define "time to development" as the expected number of years that it would take the economy to reach a given level of industrialization (measured as the

fraction of firms adopting the modern technology) from a given starting point. There are two main findings from this exercise. First, the government prefers aid in the form of a transfer, while the household is better off with a capital injection. This is because the capital injection reduces both default risk and the expected time to exit the default trap, which are future benefits that are more heavily discounted by the government. Second, debt relief not only implies the smallest increase in welfare for both the government and the household, but also has the smallest effect on time to development, and even delays development when implemented one year after the default is resolved. These results make a strong case against debt relief and in favor of aid in the form of targeted investment programs.

Layout.—Section 2 presents further evidence on the scarring effect of long defaults and on the effect of default on technology adoption, Section 3 presents the model and discusses the main novel mechanisms, Section 4 describes the calibration exercise and presents the main quantitative results, and Section 5 concludes.

2 Depressions driven by long defaults

This section analyzes how the length of default events and the depth of their real economic consequences are closely intertwined in the data. While this paper is about how long defaults may generate economic depressions, weak institutional frameworks and other underlying characteristics that predate default events may themselves jointly drive prolonged defaults and weak long-run performance. Moreover, a deep economic crisis may be the reason why a default event takes longer to resolve, with the depth of the crisis driving the duration and not the other way around.

I address these concerns in two steps. First, I use synthetic control methods to build a counterfactual for each default event, which accounts for weaknesses visible before the default. Second, I present a historical narrative of the 1980s defaults to argue that their length was determined by exogenous factors related to creditor banks and their regulators. I also argue that their duration ran well beyond what governments expected.

I also present some evidence that technology adoption drops significantly during default events and remains depressed in their aftermath, which is at the core of the main mechanism in the quantitative model that I develop in the following section.

2.1 Synthetic control analysis

In this subsection I use the synthetic control method developed by Abadie and Gardeazabal (2003) and Abadie, Diamond, and Hainmueller (2010) to measure the long-run effects of sovereign default events on output. All output data are real GDP per capita in 2011 US dollars from the Maddison Project Database (Bolt and van Zanden, 2024). To study default episodes, I use data from Reinhart and Rogoff (2009) to identify default events and measure their duration.[1]

First, I construct a synthetic time series of output in the defaulting country around each default episode. This synthetic time series serves as a proxy for the counterfactual case in which the country did not default. To illustrate the procedure, consider the case of the Argentinean default in 1982. Let $y^D_{\text{ARG},t}$ be observed real GDP per capita in Argentina in period t. I center the data so that $t = 0$ corresponds to the default year 1982. The object of interest is the unobserved time series $y^N_{\text{ARG},t}$ from $t = 0$ on, which describes how real GDP per capita in Argentina would have evolved had the government not defaulted. The synthetic control method approximates this time series as a weighted average of data from J countries in a "donor pool". In this case, the synthetic control estimators of $y^N_{\text{ARG},t}$ are

$$\hat{y}^N_{\text{ARG},t} = \sum_{j=1}^{J} \omega_j y_{j,t}, \tag{1}$$

where $y_{j,t}$ is real GDP per capita observed in country j in period t, and ω_j is the weight assigned to country j.[2] The method developed by Abadie and Gardeazabal (2003) and Abadie, Diamond, and Hainmueller (2010) chooses the vector of weights $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_J)$ so that the synthetic time series best resembles values in $T = 15$ pre-default periods:

$$\boldsymbol{\omega}^* = \arg\min_{\boldsymbol{\omega}} \left\{ \sum_{t=-T}^{-1} v_t \left(y^D_{\text{ARG},t} - \sum_{j=1}^{J} \omega_j y_{j,t} \right)^2 \right\}, \tag{2}$$

where $v_t \geq 0$ is the importance given to the fit in pre-default year t, which is chosen from the data.[3]

[1] The Reinhart and Rogoff data are annual dummies equal to one in the years that a country is in default. I merge consecutive default episodes that are separated by a single year of no default. Eight out of the 135 default events in my sample, including the longest one, are still ongoing at the end of the data, so their durations are lower bounds.

[2] These weights are restricted to be between 0 and 1 and to sum up to 1. See Abadie (2021) for a discussion about relaxing these restrictions.

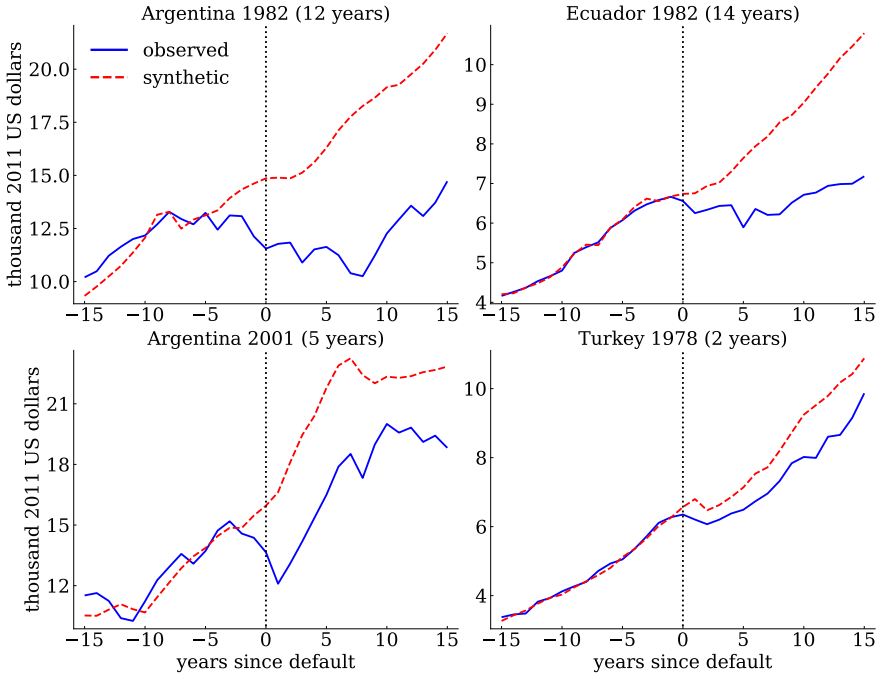
[3] I use the default settings of the `synth2` Stata package (Yan and Chen, 2023), which chooses the v_t with a

As discussed in Abadie (2021), requiring weights to be non-negative and to sum up to one typically generates sparse weight vectors (i.e., only a small number of them are non-zero). This, combined with 116 potential donor countries in both the Maddison and the Reinhart and Rogoff data, may lead to counterintuitive sets of selected donor countries. Thus, it is worth determining informed criteria to pre-select the countries that are included in the pool of J potential donors. I impose two restrictions. One is based on per-capita GDP relative to the U.S. as a proxy for development. Let this ratio for country j in period t be $x_{j,t}$ and let its average from $t = -T$ to $t = -1$ be $\bar{x}_j$. I allow country j to be in the donor pool for the Argentina-1982 case if $\bar{x}_j \in [0.6\bar{x}_{\text{ARG}}, \bar{x}_{\text{ARG}}/0.6]$. The second criterion is to only consider countries that do not default in the periods $t = -T, \ldots, 10$.[4] The top-left panel of Figure 3 presents both observed and synthetic data of real GDP per capita in Argentina around the 1982 default. The top-right panel presents the same exercise for another 1980s long default, Ecuador in 1982. The two bottom panels present the exercise for two short defaults: Argentina in 2001 and Turkey in 1978. I include Argentina because the model in Section 4 is calibrated to it, and Turkey because, like the two long defaults, its aftermath runs through the 1980s. The duration of each episode is in the title of each panel in parentheses.

regression-based procedure.

[4] Appendix B relaxes the second restriction and keeps the income band. The donor pools are then much larger and fit the pre-default data substantially better, but some donors that default inside the window are included in the construction of the "untreated" synthetic counterfactual. Any donor that defaults lowers its synthetic counterfactual and makes the estimated gap smaller. The main conclusions of the analysis, however, do not change.

Figure 3: GDP per capita around default events and their synthetic controls



Note: Observed (solid) and synthetic (dashed) real GDP per capita in thousands of 2011 US dollars. All cases are centered with the year of default at $t = 0$. The duration of each default is in parentheses.

The comparison of the top and bottom panels illustrates the deeper scarring effect of long defaults that Figure 1 suggests. As is typical with these methods, only a fraction of the potential donor countries are assigned a positive weight and the goodness of fit varies widely. Table 1 presents the countries selected by the method to construct each synthetic control and their corresponding weights. It also reports how many potential donors were considered in each case based on the two restrictions described above, and the goodness of fit in the pre-default periods measured by the root mean squared prediction error (RMSPE).

Table 1: Donor weights and fit for four synthetic controls

Argentina 1982		Ecuador 1982		Argentina 2001		Turkey 1978	
Donor	Weight	Donor	Weight	Donor	Weight	Donor	Weight
Portugal	0.779	Colombia	0.541	Colombia	0.543	Korea	0.430
New Zealand	0.221	Malaysia	0.228	Greece	0.353	Bulgaria	0.263
		Mauritius	0.214	Kuwait	0.092	El Salvador	0.251
		El Salvador	0.017	Hungary	0.012	Mauritius	0.055
Donors in pool	14		6		12		13
Pre-default RMSPE	7.55		1.06		5.70		1.69

Note: Donors with positive weight in each synthetic control. Donors in pool is the number of countries that satisfy both restrictions of income similarity and no default during the relevant window. Pre-default RMSPEs are in log points (100 times the log difference between observed and synthetic real GDP per capita).

Comparing the donor countries for the two synthetic Argentina series further illustrates the long-run transformative effect of long defaults. Before the 1982 default the method selected Portugal and New Zealand as the ones that best fit the Argentinean data. Argentina would eventually default almost twenty years later but, by then, the countries that best fit its pre-2001 data are mostly Colombia and Greece. This change reflects the decline in Argentina's per-capita GDP relative to the U.S., which was 0.47 between 1970 and 1980, on average, and 0.33 between 1990 and 2000. The cases of Ecuador 1982 and Turkey 1978 have a much better fit, as shown by lower RMSPE, and also illustrate the scarring effect of long defaults clearly. It is worth noting that Turkey's post-default period includes the same 1980s international shocks as Ecuador's (like high interest rates in the U.S.), but the post-default evolution of Turkey's GDP is much closer to its synthetic counterpart.

I perform the above analysis for all 135 defaults between 1960 and 2013 in the Reinhart and Rogoff data. I drop fourteen due to incomplete data (I only consider events with data for a $t = -15, \ldots, 20$ window) and three because the income band and no-default screen leave only one eligible donor. The remaining sample includes 118 default events with a median duration of 6 years and a maximum of 37. Table 2 presents some summary statistics from these exercises.

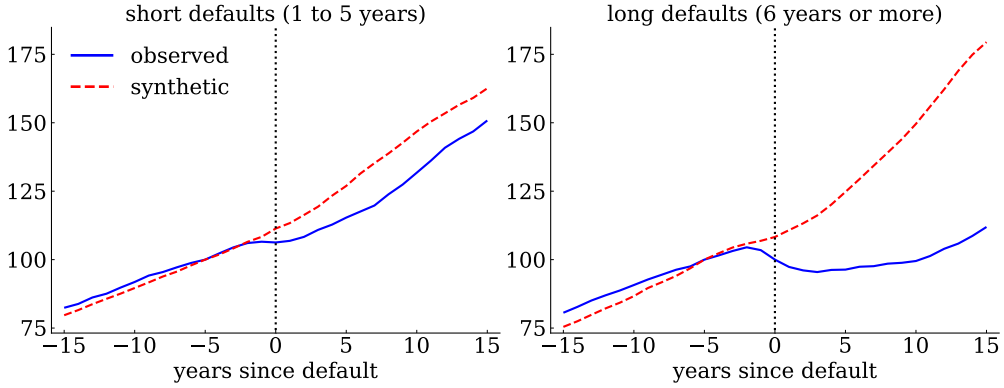
Table 2: Synthetic controls in all default events since 1960

	All	Short	Long	1980s
Default events	118	59	59	59
Default years	1960–2002	1960–2002	1965–2000	1980–1989
Median duration (years)	6	2	14	10
Duration range (years)	1–37	1–5	6–37	1–34
Median donors in pool	8	9	6	6
Median donors with positive weight	3	3	3	2
Median largest donor weight	0.60	0.59	0.62	0.64
Median pre-default RMSPE	6.72	5.29	8.20	9.18
Events not estimated	17	15	2	2
Incomplete window	14	13	1	0
Donor pool too small	3	2	1	2

Note: Pre-default RMSPEs are in log points (100 times the log difference between observed and synthetic real GDP per capita). The 1980s column includes all events that begin between 1980 and 1989. Events are not estimated if GDP data are incomplete over $t = -15, \ldots, 20$ or if fewer than two countries satisfy both restrictions of income similarity and no default during the relevant window.

I split the sample between short and long defaults at the median duration. Figure 4 presents weighted averages within each group after normalizing the year $t = -5$ to 100 (as in Figure 1). I use the reciprocal of the RMSPE of each synthetic control to construct these weighted averages so that series that are better approximated carry a higher weight (see Acemoglu et al., 2016 and Farah-Yacoub, Graf von Luckner, and Reinhart, 2024).

Figure 4: Average observed and synthetic GDP paths, short and long defaults



Note: Average observed (solid) and synthetic (dashed) real GDP per capita, normalized so that the year 5 before each default event is equal to 100. Events are weighted by the reciprocal of their pre-default RMSPE.

Long defaults have a much larger scarring effect than short ones. Note that the synthetic controls already account for idiosyncratic characteristics that are reflected in pre-default GDP, and that the synthetic economies model how the actual economy would absorb post-default aggregate

shocks had it been in good financial standing. These exercises, however, do not account for the possibility of underlying post-default confounders that may jointly drive the depth and the duration of the crises. Similarly, duration may be a result of the severity of the economic contraction and not the other way around. In the next subsection I argue that the duration of defaults in the 1980s was likely driven by exogenous factors and present evidence that these arguably exogenous lengths are related to deeper contractions in the aftermath.

2.2 Exogenously long defaults in the 1980s

I rely on the historical analysis from Almeida, Esquivel, Kehoe, and Nicolini (forthcoming), who focus on the role played by US government regulators in the buildup and aftermath of the 1980s sovereign debt crises. I summarize their analysis below.[5]

During the 1970s, oil-rich countries accumulated large amounts of savings in the banking systems of developed economies. Banks in the U.S. were encouraged by government officials to lend these funds to less-developed countries, particularly in Latin America. There is evidence that the motivation for this was political and not necessarily rational lending behavior, and that banking regulators took an abnormally lenient approach regarding the exposure that some banks acquired.[6] For example, loans to a single borrower could not exceed 10 percent of a bank's capital, but regulators allowed different agencies of the same foreign government to be considered as different borrowers. This regulatory leniency led to an overexposure of US banks to foreign sovereign borrowers. Mexico was the first country to default on its debt in 1982, followed by more than 20 other countries. For some banks, the losses generated by these defaults were almost as large as their capital. Within six months after defaulting, Mexican officials initiated attempts to restructure the debt with the banks, expecting a quick normalization of the country's financial standing. The main concern among some US regulators, however, was that such restructurings could lead to the failure of some of the exposed banks, which could create a banking panic. Thus, in order to avoid a widespread financial crisis, banks were not allowed to write off their losses until it was deemed safe to do so, which delayed the resolution of the crisis until 1990 when Mexico became the first

[5]Their analysis is based on historical accounts documented by Dooley (1995), Seidman (2000), and Silber (2012).

[6]The former head of the US Federal Deposit Insurance Corporation claims in Seidman (2000) that "The entire Ford administration, including me, told the large banks that the process of recycling petrodollars to the less developed countries was beneficial, and perhaps a patriotic duty" (p. 38).

country to restructure its debt under the Brady plan.

There are three implications of this historical evidence that are relevant for this section. First, the duration of the default events in the 1980s is likely exogenous to the fundamentals of each individual country. Had US regulators not raised systemic concerns, Mexico and its creditors would have been allowed to agree on a debt reduction as early as 1983. Fundamentals of both the country and the banks would have been reflected in the resulting restructuring, not in the duration, which was a choice made by regulators concerned with aggregate risk. Second, the main drivers of the duration of the defaults and of the depth of the contraction were not the same underlying factor. While regulatory decisions in the U.S. directly affected the length of the default episodes, they only had an indirect effect on the economic performance of the defaulting countries through their limited access to the US banking system.[7] Third, the duration of these defaults was much longer than what defaulting governments expected when they defaulted, so it could not have been anticipated in their decision to default.

Based on this historical analysis, I split the sample of default episodes into 1980s defaults and non-1980s defaults.[8] I use the synthetic economies constructed above and the following event-study design to estimate the marginal effect of default duration on economic performance:

$$\Delta y_{i,t} = \alpha_0 + \gamma_0 D_i + \sum_{k \neq 0} (\alpha_k + \gamma_k D_i) \mathbf{1}_{\{t \in B_k\}} + \varepsilon_{i,t}, \tag{3}$$

where $\Delta y_{i,t} = 100 \times \ln\left(y^D_{i,t} / \hat{y}^N_{i,t}\right)$ is the gap due to the default measured as the log difference between observed and synthetic real GDP per capita of default event i at time t, D_i is the duration of event i in years, and $\mathbf{1}_{\{t \in B_k\}}$ is the indicator function that takes the value of 1 if observation year t is in time bin B_k.[9] The five years that precede the default, $B_0 = [-5, -1]$, are the reference bin, so α_0 and γ_0 describe that bin and all remaining coefficients measure differences with respect to it. For

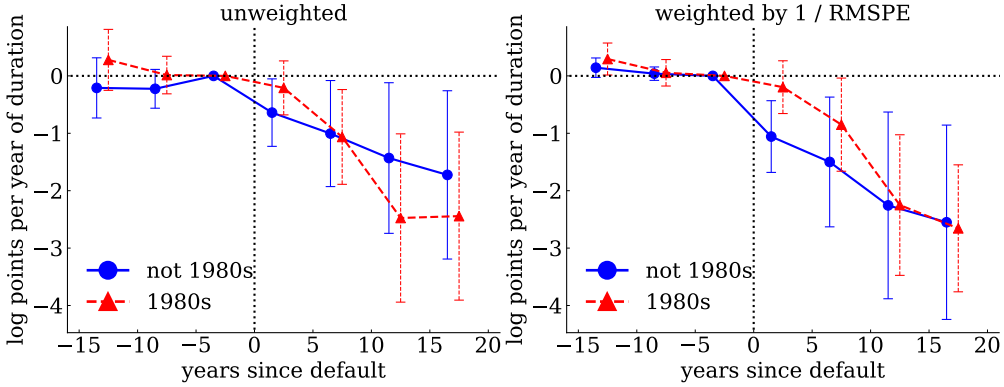
[7] The 1980s defaults also coincided with a sharp increase in US interest rates, which directly affected highly indebted countries through multiple channels (see Almeida et al., forthcoming). Note, however, that the estimated effect of duration that I present here does not follow the path of US interest rates. It is smallest in the first years after default, when rates were the highest, and largest 10 to 19 years later once rates had been lowered.

[8] The historical analysis above focuses on Latin America. Estimating equation (3) only for the 19 Latin American defaults of the 1980s gives roughly the same results.

[9] To keep the number of coefficients well below the number of default events, and still be able to analyze the effects at long horizons, I group periods into seven five-year bins: $[-15, -11]$, $[-10, -6]$, $[-5, -1]$, $[0, 4]$, $[5, 9]$, $[10, 14]$, and $[15, 19]$. Letting k correspond to single years would require estimating 72 coefficients in each sample, which is more than the 59 default events on which standard errors are clustered.

each other bin k, α_k is the intercept of the gap and γ_k is the marginal effect of one additional year of duration on the gap. I estimate equation (3) for the sample of 1980s defaults and for the sample of non-1980s defaults separately. Figure 5 presents the estimates of γ_k by weighted and ordinary least squares, together with 95 percent confidence intervals.[10]

Figure 5: The duration gradient by horizon and cohort



Note: Estimates of γ_k in equation (3) with 95 percent confidence intervals, using standard errors clustered by default event. The 1980s sample includes all events that begin between 1980 and 1989. In the right panel, events are weighted by the reciprocal of their pre-default RMSPE. The reference bin $[-5,-1]$ is set to zero.

The coefficients γ_k are negative in the years after default. Their magnitude increases with time, which shows that the cost of each additional year of default keeps growing long after the default. At long horizons, the coefficients for the 1980s defaults, whose duration was arguably driven by exogenous factors, are almost identical to those for the rest. A coefficient of -2.5 in the final bin indicates that each additional year of default is associated with a gap that is 2.5 percent deeper 15 to 19 years after the default. For an economy that experienced a 10-year default episode, the estimated gap at that horizon is 25 percent below its synthetic control.[11]

2.3 Default and technology adoption

I now present evidence that technology adoption drops significantly during default events and remains depressed years after they have been resolved. I use the data constructed by Comin and Hobijn (2009), who collect measures of the use of 104 technologies in 161 countries since 1800. I keep seventeen of these series, which measure the use of modern technologies in production,

[10] For the weighted case I use the reciprocal of the RMSPE of each synthetic control as in Figure 4.

[11] The estimated gap for an event of duration D_i in bin k is $\alpha_k + \gamma_k D_i$, relative to the five years that precede the default. In the final bin $\alpha_4 = -3.03$ and $\gamma_4 = -2.55$, so a 10-year default implies a gap of -28.5 log points, which is 25 percent below the synthetic control.

transportation and communications, and are reported by a large number of countries.[12] This is the empirical counterpart of the adoption choice in the model in Section 3, where firms choose every period whether to operate the modern technology.

I run the following regression to estimate the change in technology use during and after default events:

$$u_{c\tau t} = \alpha_{c\tau} + \lambda_{\tau t} + \kappa^D I^D_{ct} + \kappa^P I^P_{ct} + \varepsilon_{c\tau t}, \tag{4}$$

where $u_{c\tau t}$ is 100 times the log of the use of technology τ per capita in country c in year t, $\alpha_{c\tau}$ is a country-technology fixed effect, $\lambda_{\tau t}$ is a technology-year fixed effect, I^D_{ct} is an indicator that equals one if country c is in default in year t, and I^P_{ct} is an indicator that equals one if country c is in good standing in year t and was in default at some point in the previous ten years. Table 3 shows the estimated coefficients κ^D and κ^P.

Table 3: Technology use during and after default events

	(1)	(2)
In default	−14.73	−19.59
	(4.43)	(5.44)
Post-default, 10 years		−12.42
		(4.42)
Country × technology effects	Yes	Yes
Technology × year effects	Yes	Yes
Observations	57,763	57,763
Countries	93	93

Note: Standard errors are clustered at the country level.

The main results are the coefficients in column (2). During default, per capita use of these technologies drops by 18 percent and remains 12 percent lower during the ten years after a default has been resolved. In the following section I develop a quantitative sovereign default model in which endogenous technology adoption shapes long-run capital accumulation. The standard productivity cost during default makes adoption drop in the model and, if the default lasts long enough, the collapse in adoption and investment generates a scarring long-run output cost that persists well after the default is resolved.

[12] Appendix A lists them with their units and their coverage.

3 Model

There is a small open economy populated by a representative household, a government, and a continuum of firms. The household has preferences for consumption of a final good, supplies labor in a competitive market, and is excluded from international financial markets. The government has access to financial markets, and makes borrowing and investment decisions on behalf of the household. The government cannot commit to future policies and discounts the future at a higher rate than the household, which can be modeled as a result of political economy frictions (see for instance Persson and Svensson (1989), Alesina and Tabellini (1990), and Aguiar and Amador (2011)). As is standard in the sovereign default literature, the economy suffers an exogenous productivity cost while in default and is excluded from financial markets for a random number of periods. The key innovation is in the production side of the economy, which is laid out in detail below.

3.1 Environment

Technology.—There is a competitive firm that produces a final tradable good using technology

$$Y_t = \left(\int_0^1 (y_{it})^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}, \tag{5}$$

where y_{it} are differentiated non-tradable varieties of intermediate goods, and $\epsilon > 1$ is the elasticity of substitution across varieties. There is a measure one of monopolistically competitive firms $i \in [0, 1]$ that produce the intermediate goods using capital k_{it} and labor ℓ_{it}. Each intermediate good producer is characterized by an individual productivity level $z_i \geq 1$, which is fixed across time. These productivity levels are distributed according to a Pareto distribution with CDF $G(z_i) = 1 - z_i^{-\xi}$, where ξ is the tail parameter. Each period, the productivity of all firms is scaled by an aggregate productivity shock z_t, which follows a standard Markov process. This aggregate shock is the only source of risk in the model. At the beginning of each period firms choose whether to produce using a "traditional" technology

$$y_{it}^T = z_t z_i z_T k_{it}^{\alpha} \ell_{it}^{1-\alpha}, \tag{6}$$

with $\alpha \in (0,1)$, or pay a fixed cost f to operate a "modern" technology

$$y_{it}^{M} = z_t z_i z_M k_{it}^{\alpha} \ell_{it}^{1-\alpha}, \tag{7}$$

where $z_T < z_M = 1$ indicates that the modern technology is more productive. The fixed cost f is expressed in terms of the final good and has to be paid each period that the firm chooses to operate the modern technology.

Preferences and capital accumulation.—The household has preferences for consumption of the final good and labor represented by

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta_H^t \frac{\left(c_t - \frac{L_t^{1+\nu}}{1+\nu} \right)^{1-\sigma}}{1-\sigma} \right], \tag{8}$$

where $\beta_H \in (0,1)$ is the discount factor of the household, $\sigma > 0$ is the coefficient of constant relative risk aversion, and $1/\nu$ is the Frisch elasticity of labor supply. This formulation removes the wealth effect on the labor supply, preventing it from sharply rising when TFP is low or when consumption drops, which are responses that are at odds with the data (see Greenwood, Hercowitz, and Huffman, 1988). This is particularly important during default events, which are characterized by low TFP and consumption in the model. The household owns all firms and capital in the economy and rents labor and capital to the intermediate firms for a wage w_t and a rental rate r_t, respectively. The law of motion of capital is

$$K_{t+1} = I_t + (1-\delta) K_t, \tag{9}$$

where $\delta \in (0,1)$ is the depreciation rate and I_t is investment. The final good is used for investment, but there is a capital adjustment cost $\Psi(K_t, K_{t+1}) = \frac{\phi}{2} \frac{(K_{t+1}-K_t)^2}{K_t}$. This assumption, which is standard in open-economy business cycle models, prevents investment from being significantly more volatile in the model than in the data and prevents the sovereign from sharply depleting the capital stock to pay foreign debt in periods of distress, as discussed by Gordon and Guerron-Quintana (2018). As mentioned above, the government directly makes investment decisions on behalf of the household. The discussion in Subsection 3.3 below elucidates the importance of this assumption in ruling out potential sources of equilibrium multiplicity which, while interesting, are beyond the

scope of this paper.

Borrowing and default.—The government issues long-term debt in international financial markets and cannot commit to repay it. Following the literature (see Hatchondo and Martinez, 2009, Arellano and Ramanarayanan, 2012, Chatterjee and Eyigungor, 2012), I assume that debt matures at a constant rate $\gamma \in (0,1)$. The law of motion of debt is

$$B_{t+1} = i_{bt} + (1-\gamma) B_t, \tag{10}$$

where i_{bt} is current debt issuance that is sold to competitive risk-neutral international investors for a price q_t. If the government defaults then it is excluded from financial markets and gets readmitted with probability θ and all debt forgiven. While in default, aggregate productivity is $z_D(z_t) = z_t - \max\left\{0, d_0 z_t + d_1 z_t^2\right\}$, with $d_0 < 0 < d_1$. This formulation was introduced by Chatterjee and Eyigungor (2012) in a pure exchange economy and has been widely implemented in production economies as a cost to TFP (see Gordon and Guerron-Quintana, 2018 and Arellano, Bai, and Mihalache, 2018). What is important is that it makes the cost of defaulting large in "good times" when productivity is high and small or zero in "bad times", which is essential for default events and default risk to be countercyclical. Mendoza and Yue (2012) show how such an asymmetric cost of defaulting can emerge in a production economy similar to the one laid out here as a result of firms losing access to working-capital to finance purchases of certain intermediate inputs. While the above environment can be extended to feature a similarly endogenous cost, this extension would come at the expense of clarity of the novel mechanism that generates a depression after a lengthy default, which is discussed in detail below.

3.2 Aggregate production

This section characterizes the choices of individual firms and shows how increasing returns to scale arise as a result of the technology assumptions. Moreover, since all firm choices are static, this section characterizes aggregate output as a function of only the productivity shock z_t and the aggregate capital stock K_t. Crucially, the combination of (z_t, K_t) determines the mass of firms that choose to adopt the modern technology, which is a key driver of depressions following a lengthy default episode in the model.

Operating profits.—The final demand for variety i is

$$y_{it} = \left(\frac{P_t}{p_{it}}\right)^{\epsilon} Y_t, \tag{11}$$

where Y_t is output of the final good and P_t is its corresponding price index

$$P_t = \left(\int_0^1 (p_{it})^{1-\epsilon} di\right)^{\frac{1}{1-\epsilon}}. \tag{12}$$

Given a production plan y, cost minimization for firm i operating technology $j \in \{T, M\}$ yields the total cost function

$$C_{it}^j(y) = \frac{\mu_t}{z_{it}^j} y, \tag{13}$$

where $\mu_t = \left(\frac{w_t}{1-\alpha}\right)^{1-\alpha} \left(\frac{r_t}{\alpha}\right)^{\alpha}$ and z_{it}^j is firm i's productivity in period t, which depends on its individual level z_i, on the current realization of the aggregate shock z_t, and on whether the firm chose to operate the traditional technology $z_{it}^T = z_t z_i z_T$ or the modern (M) one $z_{it}^M = z_t z_i$. Given a technology choice $j \in \{T, M\}$, operating profits of firm i in period t are

$$\pi_{it}^{Oj} = \max_y \left\{ p_{it}(y)\, y - C_{it}^j(y) \right\}, \tag{14}$$

where $p_{it}(y)$ is the inverse demand function implied by (11). The solution to the problem in (14) implies the standard choice of monopoly pricing

$$p_{it}^j = \frac{\epsilon}{\epsilon - 1} \frac{\mu_t}{z_{it}^j}, \tag{15}$$

charging a markup $\frac{\epsilon}{\epsilon-1}$ over the firm's marginal cost μ_t / z_{it}^j. Plugging (15) into (11) and using the assumption that firms satisfy their demand we get that operating profits are

$$\pi_{it}^{Oj} = \frac{1}{\epsilon} \left(\frac{\epsilon - 1}{\epsilon} \frac{z_{it}^j}{\mu_t}\right)^{\epsilon-1} (P_t)^{\epsilon} Y_t. \tag{16}$$

Technology adoption.—A firm i chooses to adopt the modern technology if

$$\pi_{it}^{OM} - P_t f \geq \pi_{it}^{OT}, \tag{17}$$

that is, if the additional profits of operating the modern technology more than compensate for the fixed cost of adoption. Note that, all else equal, firms with high z_i are more profitable, so the assumption that z_i are distributed Pareto implies that there is always a positive (albeit potentially small) mass of firms that adopts the modern technology. In addition, note that firms with high z_{it}^j produce more and charge lower prices. This implies that total output Y_t increases as the mass of modern firms increases. This in turn increases demand for all varieties (equation (11)) making it more attractive for low z_i firms to adopt the modern technology.

This virtuous cycle of adoption in which some firms adopt the modern technology as other firms do so is what the development literature highlights as "the big push". Some of the work in this literature introduces such a mechanism in environments with identical firms, which gives rise to multiplicity of equilibra and generates a scope for government intervention as a coordinating force. In this model, instead, the assumptions regarding ex ante heterogeneity of firms guarantee uniqueness with respect to the adoption decision, which removes the need for coordination via government intervention. This simplifies the analysis on the interactions between increasing returns and default events, which is the main focus of the paper. Appendix (C) shows that if $(\epsilon - 1) > \frac{1+\nu}{\alpha+\nu}$ then the adoption decision is uniquely characterized by a cutoff $z_i = z_t^*$ such that equation (17) holds with equality. Under this assumption, the mass of firms adopting the modern technology $m_t = \left(\frac{1}{z_t^*}\right)^{\xi}$ is increasing in the aggregate shock z_t and in the aggregate stock of capital K_t.

TFP and final output.—Define total factor productivity as

$$A_t = \left(\int_0^1 \left(z_{it}^{j(i,t)} \right)^{\epsilon - 1} di \right)^{\frac{1}{\epsilon - 1}}, \tag{18}$$

where $j(i,t) \in \{T, M\}$ is firm i's adoption choice in period t. The assumption that z_i follow a Pareto distribution with tail parameter $\xi > \epsilon - 1$ implies that

$$A_t = A\left(z_t, z_t^*\right) = z_t \left(\frac{\xi}{1 + \xi - \epsilon} \left[1 + \left(\hat{z}^{\epsilon - 1} - 1 \right) (m_t)^{\frac{1 + \xi - \epsilon}{\xi}} \right] \right)^{\frac{1}{\epsilon - 1}}, \tag{19}$$

where z_t is the aggregate shock and m_t is the mass of firms adopting the modern technology. The condition $\xi > \epsilon - 1$ implies that the tail of the Pareto distribution cannot be too heavy (i.e. the mass of extremely productive firms should be small, and smaller for higher elasticity of substitution ϵ). The Pareto-distribution assumption allows for a closed-form characterization of A_t, and significantly simplifies the analysis and computation of equilibrium. This assumption, however, is not necessary to generate the endogenous increasing returns to scale in aggregate output. Assuming that z_i are distributed log-normal, for instance, would work similarly at the expense of having to compute A_t numerically (with a corresponding bound on the mass at the tail of the distribution).

From the market clearing conditions for production factors and using individual firm's optimality conditions we can derive total output as a function of aggregates and TFP:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}, \tag{20}$$

where L_t is the aggregate labor supply. Finally, with GHH preferences the labor supply dos not depend on the current consumption level, which makes it a static decision that can be written as a function of A_t and K_t:

$$L_t = \left[(1-\alpha) \frac{\epsilon - 1}{\epsilon} A_t K_t^{\alpha} \right]^{\frac{1}{\alpha+\nu}}. \tag{21}$$

This implies that final output is also a function of z_t and K_t only and we can define

$$F(z_t, K_t) = \left[(1-\alpha) \frac{\epsilon - 1}{\epsilon} \right]^{\frac{1-\alpha}{\alpha+\nu}} \left(A_t K_t^{\alpha} \right)^{\frac{1+\nu}{\alpha+\nu}} - P_t m_t f, \tag{22}$$

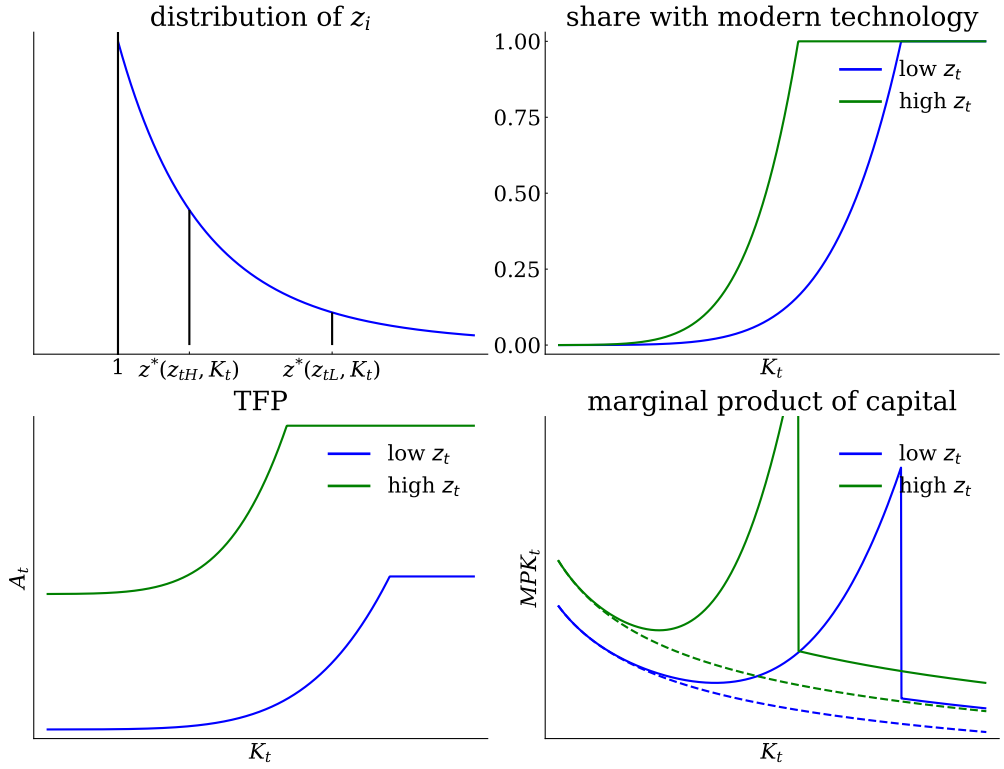
which is aggregate output net of total adoption costs (see Appendix (C) for the full derivation of these equations). These characterizations imply that the productivity cutoff implicitly defined as the solution to equation 17 is only a function of z_t and K_t

$$z_t^* = z^*(z_t, K_t) \tag{23}$$

Increasing returns to scale.—Figure 6 illustrates how capital has increasing returns to scale in some regions of the state space. The top-left panel shows the distribution of individual firm productivities z_i. There is a large mass of firms with levels of z_i close to 1, which is the lowest value,

and a small and decreasing mass of firms with high z_i. The figure also shows two productivity cutoff values corresponding to a high and low realization of the shock $z^*(z_{tH}, K_t)$ and $z^*(z_{tL}, K_t)$ for a given level of capital K_t. For the given state, all firms with productivity z_i to the right of the cutoff adopt the modern technology. As mentioned before, since the support of the Pareto distribution is $[1, \infty)$ there is always a positive mass of adopting firms. This mass, however, is lower for low realizations of z_t and close to zero for low enough levels of capital K_t, as illustrated by the top-right panel of Figure 6.

Figure 6: Capital and technology adoption



The top-right panel also shows how, for a large enough level of capital K_t all firms adopt the modern technology $m_t = 1$. The bottom-left panel shows the corresponding values of TFP A_t for high and low z_t and different levels of capital. As suggested by the behavior of m_t, TFP is low and close to being flat for low levels of capital, rapidly increasing for medium levels, and high and flat for high levels. These dynamis have important implications for the marginal product of capital MPK_t, which is shown on the bottom-right panel. For low enough levels of capital, MPK_t is decreasing in K_t, as is standard with a Cobb-Douglas production function. Once capital reaches a level at which adoption starts to quickly increase then the marginal product of capital becomes increasing as the increase in A_t more than compensates for the decreasing returns to scale in the

production function. Once capital reaches the level at which all firms adopt the modern technology $m_t = 1$, the MPK_t function features a discontinuity and becomes decreasing, but still higher than if A_t had remained constant at its level with zero adoption (illustrated by the continuing dashed lines).

3.3 Recursive formulation and equilibrium

The state of the economy is (B, K, z), where B is the debt level, K is the aggregate capital stock, and z is the aggregate productivity shock (z_t in equations (6) and (7)). At the beginning of a period in good financial standing, the value of the government is:

$$V(B,K,z) = \max_{D\in\{0,1\}} \left\{ DV^D(K,z) + (1-D)V^P(B,K,z) \right\}, \tag{24}$$

where V^D and V^P are the values of defaulting and repaying, respectively, and D is the default choice. If the government decides to repay $D = 0$, the value is

$$V^P(B,K,z) = \max_{\{B',K',c\}} \{u(c, L(z,K)) + \beta \mathbb{E}[V(B',K',z')]\} \tag{25}$$
$$s.t. \qquad c + K' + \Psi(K,K') + \gamma B \leq F(z,K) + (1-\delta)K + q(B',K',z)[B' - (1-\gamma)B]$$

where $\beta < \beta_{HH}$ is the government's discount factor, q is the price schedule for government debt (defined below), and L and F are the functions defined in the previous subsection. If the government defaults $D = 1$, the value is

$$V^D(K,z) = \max_{\{K',c\}} \left\{ u(c, L(z_D(z),K)) + \beta\theta \mathbb{E}[V(0,K',z')] + \beta(1-\theta)\mathbb{E}\left[V^D(K',z')\right] \right\} \tag{26}$$
$$s.t. \qquad c + K' + \Psi(K,K') \leq F(z_D(z),K) + (1-\delta)K,$$

where $z_D(z) = z - \max\left\{0, d_0 z + d_1 z^2\right\} \leq z$ is aggregate productivity in default.

Equilibrium.—An equilibrium is value and policy functions in default V^D, K^D, c^D; value and policy functions in repayment V, V^P, D, B^P, K^P, c^P; and a price schedule q such that: (i) given the price schedule, the value and policy functions solve the system of Bellman equations (24), (25),

and (26); and (ii) the price schedule q satisfies

$$q\left(B',K',z\right)=\frac{\mathbb{E}\left[\left(1-D\left(B',K',z'\right)\right)\left(\gamma+\left(1-\gamma\right)q\left(B'',K'',z'\right)\right)\right]}{1+r}, \tag{27}$$

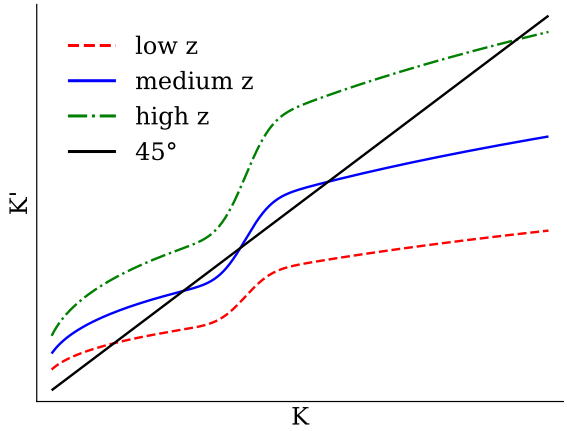
where $B''=B^{P}\left(B',K',z'\right)$, $K''=K^{P}\left(B',K',z'\right)$, and r is the international risk-free interest rate.

Uniqueness.—Before proceeding with the analysis it is important to discuss whether this dynamic equilibrium is unique or, if not, what is the scope for multiplicity. First, the conditions $\xi>\epsilon-1$ and $(\epsilon-1)>\frac{1+\nu}{\alpha+\nu}$ guarantee that there is a unique cutoff for technology adoption, which allows us to define $F\left(z_t,K_t\right)$ in equation (22). Under these assumptions the model does not feature the classic multiplicity that gives rise to coordination failures in the big-push literature, but still retains the complementarity of firms' technology adoption, which is an important mechanism from the literature. Then, the dynamic model collapses to a standard sovereign default model with production. Because the aggregate production function features increasing returns to capital there is a possibility for multiple solutions to the Euler equation for capital, such as in the model studied in Schaal and Taschereau-Dumouchel (2025). This is an important challenge in their paper because they study a decentralized environment in which competitive households make the consumption-savings decision taking prices as given. I circumvent this problem by assuming that the domestic government makes the aggregate consumption-savings choice directly. That is, the government chooses (K',B') to solve the problems in (25) and (26), taking into account how these choices affect prices and future returns (this solution can be decentralized with appropriate investment subsidies like the ones discussed in Gordon and Guerron-Quintana (2018), and corrects underinvestment of the type studied in Esquivel (2024)). If the Euler equation for capital were to have multiple solutions for a given state then the government would choose the one that yields the higher value, so the set of parameterizations under which this problem would have multiple solutions is non-generic. Finally, as shown by Aguiar and Amador (2020), sovereign default models with long-term debt may feature multiple equilibria because creditor beliefs regarding future borrowing can be self-fulfilling. To this respect, I characterize the equilibrium that results from the limit of a finite-horizon economy, following Hatchondo and Martinez (2009) and the literature thereafter.

3.4 The "Big Pull" effect of default

To analyze the dynamic implications of the production structure, consider the deterministic closed-economy version of the environment laid out above. Figure 7 illustrates the shape that the policy function for capital accumulation takes for different fixed values of the aggregate productivity shock z. For a high enough z (the green dashed-dotted line) there is a unique steady state with a high level of capital. Similarly, for a low-enough z (the red dashed line) there is a unique steady state with low capital.

Figure 7: Policy function for capital accumulation (illustrative)



The most interesting case is illustrated by the solid blue line, which features three steady-state levels of capital. The lowest and highest stead-state levels are stable, while the middle one is unstable in the sense that an epsilon-deviation from it would put the economy's capital level on a dynamic path toward one of the other steady states.

In an economy with productivity shocks, different realizations of z shift similar-looking policy functions up and down. Starting at a high enough level, capital fluctuates around the high-capital steady state for small enough shocks, but a large (and persistent) negative shock to z may push it into a trajectory toward the low-capital steady state. The converse is true for a low starting point: capital fluctuates around the low steady state and a large persistent shock may push it toward the high steady state.

This somewhat symmetric (and low-frequency) transition between fluctuations around the high and low steady states breaks in the economy with endogenous default. This is because default, which is associated with low realizations of z, triggers an additional productivity reduction through the exogenous default penalty. A lengthy default makes low values of z more persistent because

high subsequent realizations are hindered by it. When default happens in the region of the state space with high levels of capital it has a "big pull" effect on technology adoption. When the economy is in the low-capital region there is no corresponding positive event that would have a "big push" effect similar to the pull-down that default has when capital is high.

4 Quantitative analysis

I use value function iteration to solve for the equilibrium using the formulas derived in Subsection 3.2 to compute aggregate output and the labor supply as functions of the state. Following Hatchondo, Martinez, and Sapriza (2010), I compute the limit of the finite-horizon version of the economy. I jointly solve for investment and borrowing decisions using a non-linear optimization routine, and approximate value functions and the price schedule using linear interpolation. To compute expectations over the productivity shock I use a Gauss-Legendre quadrature.

4.1 Calibration

I calibrate the model to Argentina, which is a widely studied example in the sovereign default literature. There are two sets of parameters: the first (summarized in Table 4) is calibrated directly and the second (summarized in Table 5) is chosen so that moments generated by model simulations match their data counterparts. A period in the model is one quarter. The relative risk aversion parameter is $\sigma = 2$ and the risk-free rate $r^* = 0.01$, which are standard values in the business cycle literature. Following Mendoza and Yue (2012), I set $\nu = 0.455$ so that the Frisch elasticity is equal to 2.2.

Table 4: Parameters calibrated directly from the data

Parameter		Value	Source
Relative risk aversion	σ	2	Standard value
Risk-free rate	r^*	0.01	Standard value
Labor disutility parameter	ν	0.455	Mendoza and Yue (2012)
Debt duration	γ	0.05	Chatterjee and Eyigungor (2012)
Probability of reentry	θ	0.0625	Gelos, Sahay, and Sandleirs (2011)
Capital share	α	0.33	Standard value
Elasticity of substitution	ϵ	2.92	Schaal and Taschereau-Dumouchel (2024)
Depreciation rate	δ	0.05	Standard value
Mean of productivity shock	μ_z	0.768	High-capital steady state $y^H_{ss} = 1$
Persistence of productivity	ρ	0.95	Gordon and Guerron-Quintana (2018)
Variance of productivity	σ^2	0.017	Gordon and Guerron-Quintana (2018)
Pareto curvature	ξ	20.5	high-capital steady state with $m_t = 1$
Fixed cost	f	0.04	low-capital steady state with $m_t < 0.1$
Productivity of traditional technology	z_T	0.898	$y^L_{ss}/y^H_{ss} = 0.60$

I set the debt duration parameter to $\gamma = 0.05$ to match the maturity information for Argentina reported in Broner, Lorenzoni, and Schmukler (2013), as done by Chatterjee and Eyigungor (2012). I set the probability of reentry to financial markets to $\theta = 0.0625$ for an average duration in autarky of 16 quarters, which is the median duration of default events documented by Gelos, Sahay, and Sandleirs (2011). On the production side, I set the elasticity of substitution to $\epsilon = 2.92$, following Schaal and Taschereau-Dumouchel (2024) who calibrate a similar model of heterogeneous firms. This is close to the value of 3, which is standard in the literature (see the discussion in Hsieh and Klenow (2009)). The capital share $\alpha = 0.33$ and the capital depreciation rate $\delta = 0.05$ are standard values. The aggregate productivity shock follows an AR(1) process

$$\log z_t = (1-\rho)\log \mu_z + \rho \log z_{t-1} + \sigma_z \epsilon_t, \tag{28}$$

where the persistence $\rho = 0.95$ and variance $\sigma^2 = 0.017$ are taken from Gordon and Guerron-Quintana (2018), who calibrate them for Argentina using a Cobb-Douglas production technology. I set the mean value of the shock $\mu_z = 0.768$ so that output $y^H_{ss} = 1$ in the deterministic steady-state in which all firms adopt the modern technology. Let y^L_{ss} be the level of output in the hypothetical steady state in which no firms adopt the modern technology, I set the productivity level of the traditional techology $z_T = 0.898$ so that $y^L_{ss} = 0.6$, which is the average drop of output from trend observed in depressions that follow lengthy defaults. I jointly choose the Pareto tail parameter

$\xi = 20.5$ and the fixed cost of adoption $f = 0.04$ so that $m_t = 1$ at the steady-state level of capital with full adoption k_{ss}^H and so that $m_t < 0.1$ at the hypothetical steady-state level of capital with no adoption k_{ss}^L. These choices will allow the ergodic distribution to feature long periods of high and low technology adoption.

Finally, I set the discount factor $\beta = 0.9697$, the parameters governing the default penalty to productivity $d_0 = -0.4826$ and $d_1 = 0.5882$, and the capital adjustment cost $\phi = 3.1395$, to jointly match average spreads of 0.08, an average debt-to-GDP ratio of 0.31, a default probability of 0.03, and a volatility of investment of 12.8. I interpret the Argentinean data as being generated by an economy that is fluctuating around a low-adoption long-run steady state, which is consistent with high default risk and low GDP in the model. To that end, the targeted moments in the model are calculated conditional on the fraction of firms adopting the modern technology being below its long-run median $m_t \leq 0.17$. Table 5 summarizes this calibration exercise.

Table 5: Parameters calibrated simulating the model

Parameter		Value	Moment	Data	Model, low adoption
discount factor	β	0.9697	default probability	0.03	0.032
default cost	d_0	-0.4826	$Av\,(r - r^*)$	0.08	0.08
default cost	d_1	0.5882	$Av\left(\frac{b}{gdp}\right)$	0.31	0.31
capital adjustment cost	ϕ	3.1395	σ_i	12.8	11.5

To compute the model moments I draw 100 samples of 11,000 periods and drop the first 1,000. Since the targeted moments are conditional on low adoption, I also drop all periods in which the mass of adopters is below its unconditional median. Investment data is HP-Filtered with a smoothing parameter of 1,600. In the data, debt corresponds to External debt stocks, public and publicly guaranteed from the World Development Indicators database. All model moments are conditional on the economy being in good financial standing.

Table 6 compares other business cycle statistics from the model to their data counterparts. Column (2) in Table 6 reports moments conditional on m_t being below its median, as in the calibration exercise. Column (3) reports them conditional on m_t being above its median and Column (4) reports them conditional on full adoption $m_t = 1$.

Table 6: Business cycle statistics

			model	
	(1)	(2)	(3)	(4)
moment	data	low adoption	high adoption	full adoption
Default frequency	0.03	0.016	0.014	0.008
Av. spreads	0.08	0.08	0.05	0.04
Std. spreads	0.04	0.12	0.06	0.03
debt-to-GDP ratio	0.31	0.31	0.38	0.38
Capital stock (total)	n.a.	2.64	4.24	4.71
σ_c/σ_y	1.23	1.01	0.99	0.87
σ_i/σ_y	2.65	2.24	2.06	1.64
$\sigma_{TB/y}$	2.34	4.38	4.02	3.23
$Cor\,(\text{spreads}, y)$	-0.79	-0.17	-0.35	-0.48
$Cor\,(TB/y, y)$	-0.68	-0.11	-0.13	-0.16

To compute the model moments I draw 100 samples of 11,000 periods and drop the first 1,000. Column (2) drops all periods in which the mass of adopters is above its unconditional median, Column (3) drops all periods in which the mass of adopters is below the median, and Column (4) drops all the periods in which the mass of adopters is not 1. Except for spreads and the debt-to-GDP ratio, all variables are HP-Filtered with a smoothing parameter of 1,600. In the data, debt corresponds to External debt stocks, public and publicly guaranteed from the World Development Indicators database. All model moments are conditional on the economy being in good financial standing.

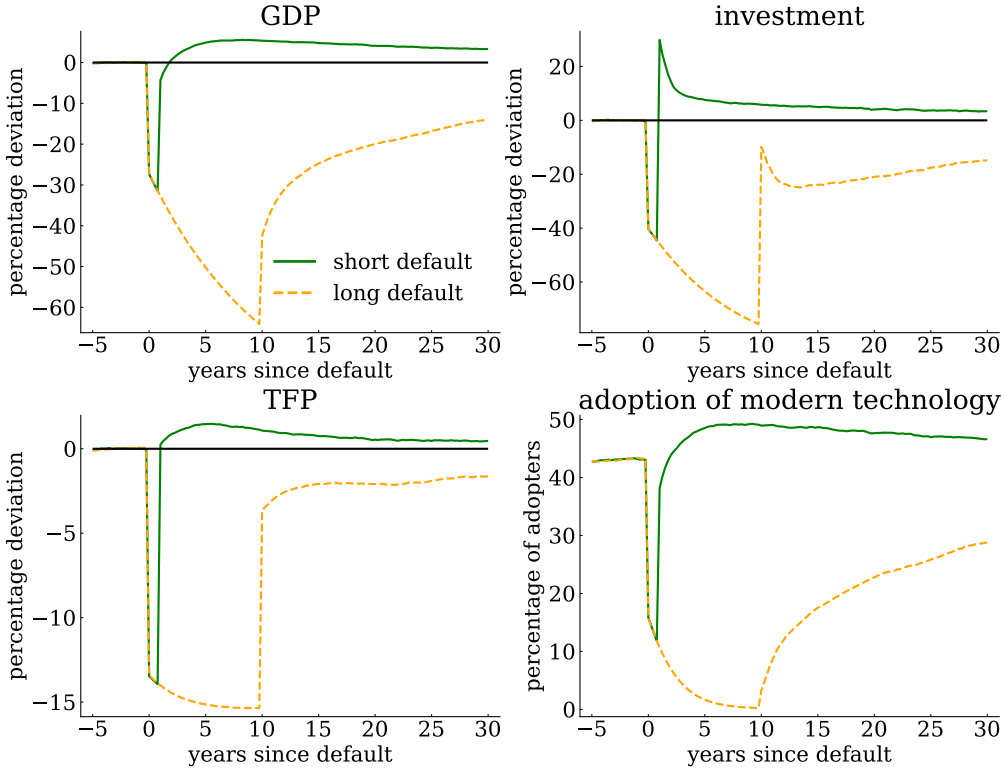
The model does a good job of replicating long-run business cycle statistics of Argentina during low adoption periods, including moments that were not directly targeted by the calibration exercise. Columns (3) and (4) show how periods of high adoption are associated with higher capital accumulation, lower default risk, and lower spreads, which allow the economy to support higher debt levels (it is worth noting that, while the median adoption level is well below 1, there is full adoption 34 percent of the time). In contrast, the region of the ergodic distribution with low adoption features spreads that are significantly higher and more volatile, the aggregate capital stock is roughly half of what it is in periods with high adoption, and the economy is not able to sustain as much debt. Once the economy gets pulled down to this part of the ergodic distribution, more frequent defaults (even if they are short) make transitioning back to normal times less likely. This point is explored further in the following subsections.

4.2 Default depressions

To analyze the different effects of short and long defaults in the model I perform the following two exercises. First, I consider a random draw from the ergodic distribution and exogenously shift the economy to the default state, regardless of whether the government would have chosen to default

or not. I then impose default durations of one year and ten years, and continue the simulation for 30 years after the induced default. Figure 8 shows the average of 10,000 of the paths for each default duration (one year duration is the green solid line and ten year duration is the yellow dashed line), between five years before and 30 years after the default began.

Figure 8: Exogenously induced default, short and long



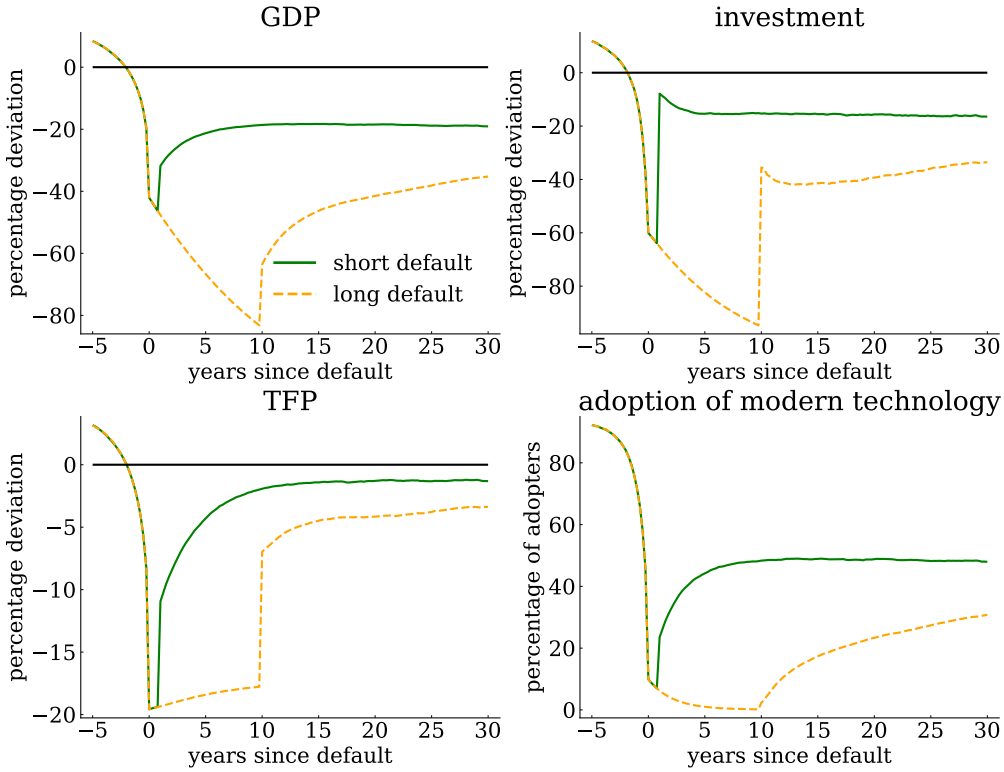
Each line is the average of 10,000 paths for each default duration (one year duration is the green solid line and ten year duration is the yellow dashed line). The paths are centered around an exogenously induced default in year $t = 0$.

The average drop in GDP is initially the same, but its recovery in the long-run is significantly different, just like in the data presented in Figure 1 in the introduction. The top-left panel shows that GDP bounces back after the short default, driven by the recovery in technology adoption that increases TFP. GDP after the long default event also recovers but to a much lower level. Moreover, GDP in this case remains well below its pre-default level 20 years after the default crisis ended. This is because the stock of capital falls so much that technology adoption does not recover after the crisis ends and TFP remains at a much lower level.

Figure 9 does a similar exercise for the typical default in the model that happens following periods with full-adoption. That is, I now select defaults that happen endogenously in the model following periods with full adoption and impose the same two alternative durations. As is standard

in these models, default episodes are preceded by a sequence of bad shock realizations, which explain why GDP, investment, TFP, and technology adoption are already dropping in the years prior.

Figure 9: Typical default event, short and long



Each line is the average of 10,000 paths for each default duration (one year duration is the green solid line and ten year duration is the yellow dashed line). The paths are centered around an endogenous default in year $t = 0$ that followed a period with full technology adoption.

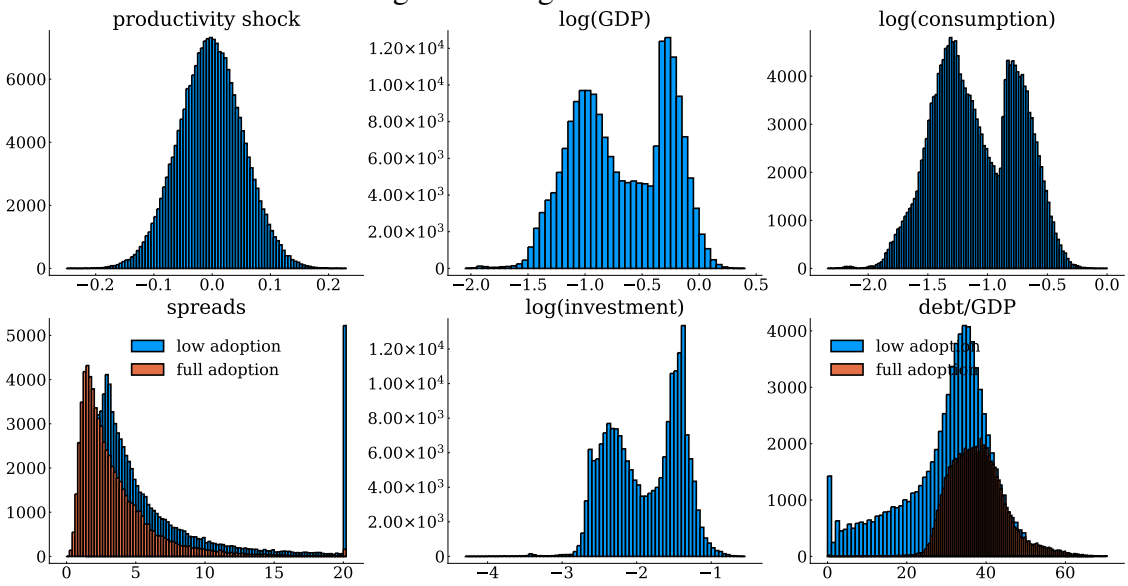
As before, the duration of the default episodes matters. GDP, TFP, investment, and adoption all bounce back to a higher level in the economy with the short default episode than in the economy with the long default. The long default has a different and long-lasting negative effect that remains 20 years after the event was resolved. In general, average adoption of the modern technology does not fully recover after default, increasing the vulnerability of the economy to future negative shocks even after short defaults.

4.3 The default trap

As suggested by Table 6 and consistent with the previous analysis regarding two stable steady states, Figure 10 shows that the ergodic distribution of the endogenous variables in the model is

bimodal. It presents histograms for the natural logarithm of the productivity shock (relative to its mean), the natural logarighm of GDP, consumption, and investment from 200,000 periods simulated from the model. It also presents two conditional histograms for the debt-to-GDP ratio and spreads: one with full adoption ($m_t = 1$) and one with adoption below the median (the histograms for spreads feature an overflow bin to the right for values ≥ 20).

Figure 10: Ergodic distribution



Each histogram is constructed from a sample of 200,000 periods simulated from the ergodic distribution of the model. The panels for spreads and debt each present two conditional histograms: one with full adoption ($m_t = 1$) and one with adoption below the median (the histograms for spreads feature an overflow bin to the right for values ≥ 20).

The histograms for GDP, consumption, and investment are not only bimodal, but accumulate more mass on the lower end of the distribution. As can be seen in columns (3) and (4) of Table 6, when capital is abundant technology adoption is high and defaults are infrequent. This makes the cost of defaulting relatively high because there is a probability that, if the default lasts long enough, the economy will be pulled down to the "poor" side of the ergodic distribution. With a relatively high cost of defaulting the economy sustains higher average levels of debt and lower spreads, illustrated by the orange histograms in the bottom-right and bottom-left panels, respectively.

When capital is scarce (column (2) in Table 6), this component of the cost of defaulting is absent, making default more attractive and spreads higher. The spreads distribution of the low-adoption periods is to the right of that of the high adoption and has a fatter tail. In addition, the government has an additional incentive to accumulate debt rapidly besides its preference for front-loading consumption. When the stock of capital is close to the region with increasing returns,

borrowing to invest becomes attractive despite the large cost from high spreads because the economy may get into a path toward the "rich" side of the ergodic distribution. This is a gamble that may not pay off given the high probability of a set-back due to a default along the way. This creates the default trap in the model.

4.4 Foreign aid

The model studied in this paper has little scope for the government to intervene and improve economic outcomes. In particular, there is no scope for a big push by increasing borrowing from abroad to boost demand and industrialization. The domestic government already has instruments to select all aggregate allocations, and the big pull around the default trap stems from the government's inability to commit to future policy and its bias toward front-loading consumption.

I now explore the effects of foreign aid following a ten-year default. The main objective is to evaluate the welfare and development implications of different types of aid. The experiment considers a one-time transfer of resources a from an international entity in the form of a non-defaultable perpetuity that has to be serviced every period at a subsidized interest rate $r_a < r$. The resource constraint in repayment after receiving aid is now

$$c + K' + \Psi(K, K') + \gamma B + r_a a \leq F(z, K) + (1-\delta) K + q(B', K', z)\left[B' - (1-\gamma) B\right],$$

and in default it is

$$c + K' + \Psi(K, K') + r_a a \leq F(z_D(z), K) + (1-\delta) K.$$

I consider three types of aid: an unconditional transfer a that can be freely allocated by the government, debt relief so that $B' = B - a$, and a capital injection so that $K' = K + a$. I compute welfare gains in consumption equivalent units for the household and the government. To measure the effect on development, I compute the following "time to development" variable

$$\mathcal{T}_0(\tilde{m}) = \mathbb{E}\left[\text{years until } m_t = \tilde{m} \,|\, (z_0, K_0, B_0)\right]$$

which is the expected number of years from period 0, given a state (z_0, K_0, B_0), before the economy reaches a mass $\tilde{m}$ of firms that adopts the modern technology. I set a to be equal to 15 percent of

average GDP, $r_a = r/2$, and $\beta_{HH} = \frac{1}{1+r}$. I compute the time to development for $\tilde{m} = 0.5$ and for full adoption $\tilde{m} = 1.0$. Table 7 presents the results.

Table 7: Effects of foreign aid on welfare and development

case		welfare gains: government (1)	welfare gains: household (2)	m_0 (3)	time to development: $\tilde{m} = 0.5$ (4)	time to development: $\tilde{m} = 1.0$ (5)
1 year after default resolution	no aid	0.00	0.00	0.11	42	52
	debt relief	3.81	2.22	0.11	44	54
	capital injection	3.82	5.06	0.11	37	47
	transfer	5.08	3.36	0.11	36	46
5 years after default resolution	no aid	0.00	0.00	0.17	38	47
	debt relief	3.98	2.38	0.17	35	45
	capital injection	4.24	5.38	0.17	33	42
	transfer	5.51	3.54	0.17	33	42

The first four rows present the results of implementing the aid program one year after the default has been resolved, and the next four correspond to implementing the program 5 years later. Column (3) shows that one year after the resolution of the default episode 11 percent of firms adopt the modern technology, while 17 percent do so five years later. In both cases welfare gains for the government are the highest if the program is an unconditional transfer. This would be the optimal policy if the government were benevolent, since it is a non-distortionary lump-sum transfer. Note, however, that the household strictly prefers the capital injection because it reduces default risk and the expected time to exit the default trap, both of which are future benefits that are more heavily discounted by the impatient government. Both the government and the household prefer capital injections over debt relief, regardless of the timing of aid. Moreover, debt relief not only implies the smallest increase in welfare, but also has the smallest effect on pushing the economy out of the default trap, and even delays development when implemented one year after the default is resolved. These results make a strong case against debt relief and in favor of aid in the form of targeted investment programs.

5 Conclusion

Motivated by the effects of long defaults in the data, this paper presents a novel theory of default-induced depressions. In the model, long default episodes pull the economy down to a default trap in which the capital stock and technology adoption are lower, defaults are more frequent, and

spreads are higher and more volatile than in normal times. Besides these novel dynamics, the model features all the standard business cycle properties of emerging economies that face sovereign risk, as all the workhorse models in this literature. The model allows for the evaluation of different forms of foreign aid and makes a strong case against debt relief, favoring aid in the form of capital injections instead. This is because capital injections relax the resource constraint and push the economy closer to a development path, while debt relief only achieves the former. This difference is stronger under political economy frictions that make the government relatively more impatient than its constituent households.

The model sheds light on a plausible mechanism through which an emerging (or even an advanced) economy may become trapped in a low-income and high-default risk regime. The results from the model and the motivating evidence stress the importance of understanding the sources of lengthy default episodes and different policies to prevent them. It is tempting to conclude that the model makes a strong case for swift resolutions to sovereign crises through the intervention of international financial institutions. Moreover, one may conclude that this case is even stronger for crises in rich economies, who stand a lot more to lose by falling into a default trap. The effects of such interventions, however, are not innocuous because of the potential ex ante effects they may have on borrowing and default incentives. This scope for moral hazard would also have significant consequences on the regime switching dynamics introduced in this paper. Better understanding the effects of such interventions and other similar policy proposals is an exciting avenue for future research.

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A Technology series

Table 8 lists the seventeen series from the CHAT dataset that I use in the regression reported in Table 3. The years and country counts are those of the estimation sample. I drop technology-year cells with fewer than ten reporting countries, and restrict the panel to countries whose default status is known.

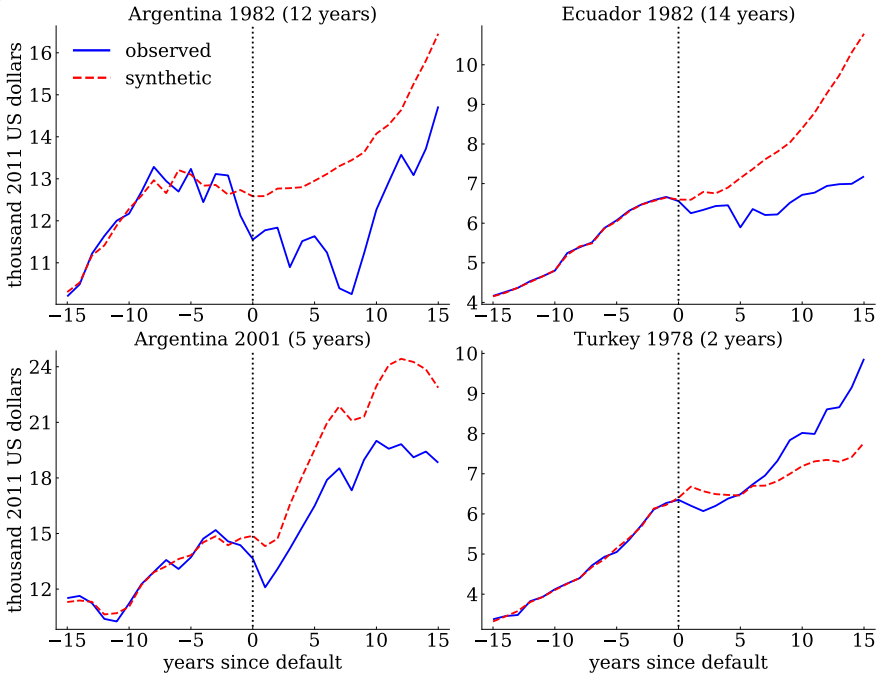
Table 8: Technology series used in the adoption exercise

Technology	Years	Countries	Units
Telegrams sent	1865–1993	57	millions
Rail freight traffic	1888–1993	72	million tonne-km
Rail passenger traffic	1891–1993	66	million passenger-km
Telephone lines in use	1900–2002	93	count
Electricity production	1912–2001	91	kWh
Passenger cars in use	1920–2001	93	count
Commercial vehicles in use	1920–2001	85	count
Air passenger traffic	1929–1993	76	million passenger-km
Radio receivers	1933–1999	93	thousands
Steel, all processes	1936–2001	65	thousand metric tons
Air freight traffic	1945–1991	72	million tonne-km
Television sets	1953–2002	93	count
Fertilizer consumption	1961–2001	93	metric tons
Tractors in use	1961–2001	93	count
Mobile subscribers	1985–2002	92	count
Personal computers	1988–2002	87	count
Internet users	1990–2002	93	count

B Synthetic controls with the relative-income restriction only

XXX

Figure 11: GDP per capita around default events and their synthetic controls (relative-income restriction only)



Note: XXX.

XXX

Table 9: Donor weights and fit for four synthetic controls (relative-income restriction only)

Argentina 1982		Ecuador 1982		Argentina 2001		Turkey 1978	
Donor	Weight	Donor	Weight	Donor	Weight	Donor	Weight
Iran	0.284	Malaysia	0.418	Venezuela	0.459	Bolivia	0.336
Chile	0.273	Uruguay	0.113	Panama	0.284	Romania	0.223
New Zealand	0.195	Bolivia	0.096	Uruguay	0.163	Korea	0.142
Venezuela	0.136	Mauritius	0.095	Kuwait	0.094	Bulgaria	0.097
Finland	0.105	Poland	0.074			Iraq	0.062
United Kingdom	0.007	Brazil	0.065			Seychelles	0.047
		Cuba	0.051			Singapore	0.045
		Jamaica	0.047			Gabon	0.030
		Nicaragua	0.013			Mauritius	0.018
		Peru	0.011				
		Paraguay	0.008				
		Iraq	0.004				
		Jordan	0.003				
Donors in pool	28		34		28		35
Pre-default RMSPE	2.38		0.37		2.20		1.11

Note: XXX.

XXX

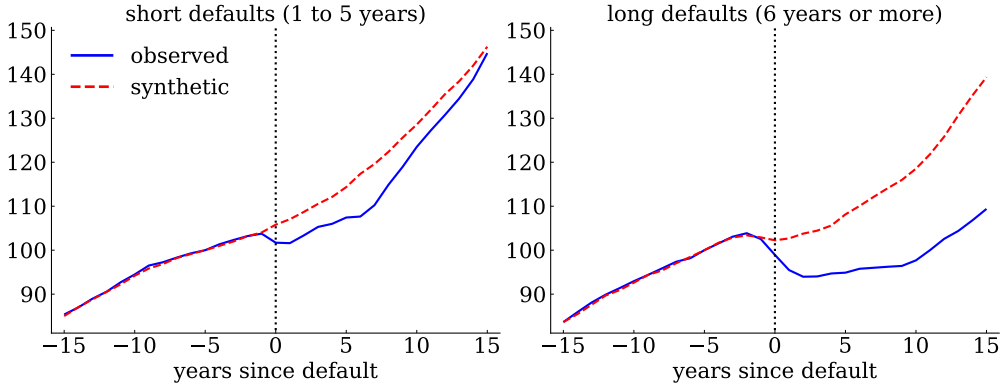
Table 10: Synthetic controls in all default events since 1960 (relative-income restriction only)

	All	Short	Long	1980s
Default events	121	61	60	61
Default years	1960–2002	1960–2002	1965–2000	1980–1989
Median duration (years)	5	2	14	10
Duration range (years)	1–37	1–5	6–37	1–34
Median donors in pool	33	32	33	33
Median donors with positive weight	6	6	6	6
Median largest donor weight	0.41	0.40	0.41	0.40
Median pre-default RMSPE	2.62	2.59	2.64	2.62
Events not estimated	14	13	1	0
Incomplete window	14	13	1	0
Donor pool too small	0	0	0	0

Note: XXX.

XXX

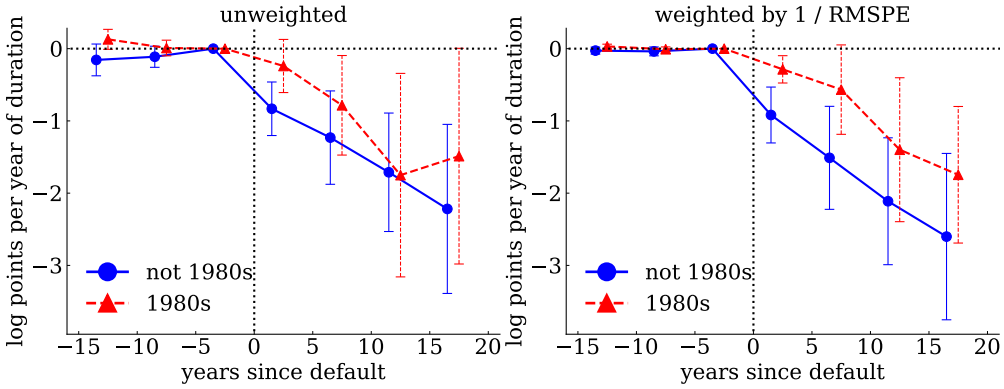
Figure 12: Average observed and synthetic GDP paths, short and long defaults (relative-income restriction only)



Note: XXX.

XXX

Figure 13: The duration gradient by horizon and cohort (relative-income restriction only)



Note: XXX.

C Characterization of the adoption cutoff

C.1 Problem of a firm

- Given a production plan y_{it}, the cost minimization problem of the firm is

$$\min_{k_{it},l_{it}} r_t k_{it} + w_t l_{it}$$

$$s.t. \qquad y_{it} \leq z_{it} k_{it}^{\alpha} l_{it}^{1-\alpha}$$

the F.O.C.s are

$$r_t = \eta_{it} \alpha \frac{z_{it} k_{it}^{\alpha} l_{it}^{1-\alpha}}{k_{it}}$$

$$w_t = \eta_{it} (1-\alpha) \frac{z_{it} k_{it}^{\alpha} l_{it}^{1-\alpha}}{l_{it}}$$

where $\eta_{i,t}$ is the multiplier of the constraint. Assume the constraint binds so $\eta_{it} > 0$, rearranging we get

$$\frac{k_{it}}{l_{it}} = \frac{\alpha}{1-\alpha} \frac{w_t}{r_t}$$

plugging into the constraint we get

$$l_{it} = \left(\frac{1-\alpha}{\alpha}\frac{r_t}{w_t}\right)^{\alpha}\frac{y_{it}}{z_{it}}$$
$$k_{it} = \left(\frac{\alpha}{1-\alpha}\frac{w_t}{r_t}\right)^{1-\alpha}\frac{y_{it}}{z_{it}}$$

so the cost function is

$$\begin{aligned} C_{it}(y_{it}) &= \left(\frac{w_t}{1-\alpha}\right)^{1-\alpha}\left(\frac{r_t}{\alpha}\right)^{\alpha}\frac{y_{it}}{z_{it}} \\ &= \frac{\mu_t}{z_{it}}y_{it} \end{aligned}$$

where $\mu_t = \left(\frac{w_t}{1-\alpha}\right)^{1-\alpha}\left(\frac{r_t}{\alpha}\right)^{\alpha}$

- Operating profits are

$$\pi_{it}^{O} = \max_{y_{it}} p_{it}(y_{it})\, y_{it} - \frac{\mu_t}{z_{it}}y_{it}$$

using the demand curve the F.O.C. implies

$$p_{it} = \frac{\epsilon}{\epsilon-1}\frac{\mu_t}{z_{it}}$$
$$y_{it} = \left(\frac{P_t}{p_{it}}\right)^{\epsilon}Y_t$$

price is a markup $\frac{\epsilon}{\epsilon-1}$ over marginal cost μ_t/z_{it}. Operating profits are then

$$\begin{aligned} \pi_{it}^{O} &= \frac{1}{\epsilon-1}\frac{\mu_t}{z_{it}}y_{it} \\ &= \frac{1}{\epsilon}\left(\frac{\epsilon-1}{\epsilon}\frac{z_{it}}{\mu_t}\right)^{1-\epsilon}(P_t)^{\epsilon}Y_t. \end{aligned}$$

C.2 Price of final good and aggregate productivity

- The price of the final good satisfies

$$\begin{aligned} P_t^{1-\epsilon} &= \int_0^1 (p_{it})^{1-\epsilon}\, di \\ &= \int_0^1 \left(\frac{\epsilon}{\epsilon-1}\frac{\mu_t}{z_{it}}\right)^{1-\epsilon} di \\ &= \left(\frac{\epsilon-1}{\epsilon}\frac{1}{\mu_t}\right)^{\epsilon-1} \int_0^1 (z_{it})^{\epsilon-1}\, di \end{aligned}$$

  define a productivity aggregator as

$$A_t = \left(\int_0^1 (z_{it})^{\epsilon-1}\, di\right)^{\frac{1}{\epsilon-1}}$$

  then the price of the final good is

$$P_t = \frac{\epsilon}{\epsilon-1}\frac{\mu_t}{A_t}$$

- Note that

$$\int_0^1 (z_{it})^{\epsilon-1}\, di = \int_1^\infty (z_{it})^{\epsilon-1}\, dF(z_i)$$

  let the break-even cutoff productivity to operate the modern technology be z_t^*, then

$$\int_0^1 (z_{it})^{\epsilon-1}\, di = \int_1^{z_t^*} (z_{it})^{\epsilon-1}\, dF(z_i) + \int_{z_t^*}^\infty (z_{it})^{\epsilon-1}\, dF(z_i)$$

  recall that z_i are distributed Pareto, so

$$\begin{aligned} F(z_i) &= 1 - z_i^{-\xi} \\ dF(z_i) &= \xi z_i^{-\xi-1} \end{aligned}$$

  plugging in and rearranging

$$\int_0^1 (z_{it})^{\epsilon-1}\, di = \xi\,(\hat{z} z_t)^{\epsilon-1} \int_1^{z_t^*} (z_i)^{\epsilon-\xi-2}\, dz_i + \xi\,(z_t)^{\epsilon-1} \int_{z_t^*}^\infty (z_i)^{\epsilon-\xi-2}\, dz_i$$

which implies that the integrals are

$$\begin{aligned}\int_1^{z_t^*} (z_i)^{\epsilon-\xi-2}\, dz_i &= \frac{(z_i)^{\epsilon-\xi-1}}{\epsilon-\xi-1}\Big|_1^{z_t^*} \\ &= \frac{1}{1+\xi-\epsilon}\left[1-\left(\frac{1}{z_t^*}\right)^{1+\xi-\epsilon}\right] \\ \int_{z_t^*}^{\infty} (z_i)^{\epsilon-1}\, dF(z_i) &= \frac{(z_i)^{\epsilon-\xi-1}}{\epsilon-\xi-1}\Big|_{z_t^*}^{\infty} \\ &= \frac{1}{1+\xi-\epsilon}\left(\frac{1}{z_t^*}\right)^{1+\xi-\epsilon}\end{aligned}$$

and note that the second is well defined iff $0 < 1+\xi-\epsilon$. Plugging in and rearranging

$$\int_0^1 (z_{it})^{\epsilon-1}\, di = (z_t)^{\epsilon-1}\frac{\xi}{1+\xi-\epsilon}\left[z_T^{\epsilon-1}+\left(1-z_T^{\epsilon-1}\right)\left(\frac{1}{z_t^*}\right)^{1+\xi-\epsilon}\right]$$

- So the productivity aggregator is

$$A_t = A\left(z_t^*\right) = z_t\left(\frac{\xi}{1+\xi-\epsilon}\left[z_T^{\epsilon-1}+\left(1-z_T^{\epsilon-1}\right)\left(\frac{1}{z_t^*}\right)^{1+\xi-\epsilon}\right]\right)^{\frac{1}{\epsilon-1}}$$

note that in the limit where no firm adopts the modern technology we get

$$\lim_{z_t^*\to\infty} A\left(z_t^*\right) = z_t z_T\left(\frac{\xi}{1+\xi-\epsilon}\right)^{\frac{1}{\epsilon-1}}$$

and in the limit where all firms adopt the modern technology we get

$$\lim_{z_t^*\to 1} A\left(z_t^*\right) = z_t\left(\frac{\xi}{1+\xi-\epsilon}\right)^{\frac{1}{\epsilon-1}}$$

C.3 Aggregate production

- Taking stock, we know:

$$
\begin{aligned}
l_{it} &= \left(\frac{1-\alpha}{\alpha}\frac{r_t}{w_t}\right)^{\alpha}\frac{y_{it}}{z_{it}} \\
k_{it} &= \left(\frac{\alpha}{1-\alpha}\frac{w_t}{r_t}\right)^{1-\alpha}\frac{y_{it}}{z_{it}} \\
\mu_t &= \left(\frac{w_t}{1-\alpha}\right)^{1-\alpha}\left(\frac{r_t}{\alpha}\right)^{\alpha} \\
p_{it} &= \frac{\epsilon}{\epsilon-1}\frac{\mu_t}{z_{it}} \\
y_{it} &= \left(\frac{P_t}{p_{it}}\right)^{\epsilon} Y_t \\
p_{it}y_{it} &= (p_{it})^{1-\epsilon}(P_t)^{\epsilon} Y_t \\
A_t &= \left(\int_0^1 (z_{it})^{\epsilon-1}\, di\right)^{\frac{1}{\epsilon-1}} \\
P_t &= \frac{\epsilon}{\epsilon-1}\frac{\mu_t}{A_t}
\end{aligned}
$$

- Combining individual factor demands with optimal production plans and rearranging we get

$$
\begin{aligned}
l_{it} &= \frac{1-\alpha}{w_t}\frac{\epsilon-1}{\epsilon}(p_{it})^{1-\epsilon}(P_t)^{\epsilon} Y_t \\
k_{it} &= \frac{\alpha}{r_t}\frac{\epsilon-1}{\epsilon}(p_{it})^{1-\epsilon}(P_t)^{\epsilon} Y_t
\end{aligned}
$$

  integrating both sides over all firms i

$$
\begin{aligned}
\int_0^1 l_{it}di &= \int_0^1 \frac{1-\alpha}{w_t}\frac{\epsilon-1}{\epsilon}(p_{it})^{1-\epsilon}(P_t)^{\epsilon} Y_t di \\
\int_0^1 k_{it}di &= \int_0^1 \frac{\alpha}{r_t}\frac{\epsilon-1}{\epsilon}(p_{it})^{1-\epsilon}(P_t)^{\epsilon} Y_t di
\end{aligned}
$$

simplifying and using market clearing

$$L_t = \frac{1-\alpha}{w_t}\frac{\epsilon-1}{\epsilon}P_t Y_t$$
$$K_t = \frac{\alpha}{r_t}\frac{\epsilon-1}{\epsilon}P_t Y_t$$

so we get the aggregate capital-to-labor ratio is

$$\frac{K_t}{L_t} = \frac{\alpha}{1-\alpha}\frac{w_t}{r_t} = \frac{k_{it}}{l_{it}} \qquad \forall i$$

- Production of firm i is

$$y_{it} = \left(\frac{P_t}{p_{it}}\right)^{\epsilon} Y_t$$

plugging in for p_{it} and P_t we get

$$y_{it} = \left(\frac{\frac{\epsilon}{\epsilon-1}\frac{\mu_t}{A_t}}{\frac{\epsilon}{\epsilon-1}\frac{\mu_t}{z_{it}}}\right)^{\epsilon} Y_t$$
$$= \left(\frac{z_{it}}{A_t}\right)^{\epsilon} Y_t$$

plugging in for the production function and rearranging

$$\left(\frac{k_{it}}{l_{it}}\right)^{\alpha} l_{it} = (z_{it})^{\epsilon-1}\left(\frac{1}{A_t}\right)^{\epsilon} Y_t$$

plugging in for the aggregate capital-to-labor ratio

$$\left(\frac{K_t}{L_t}\right)^{\alpha} l_{it} = (z_{it})^{\epsilon-1}\left(\frac{1}{A_t}\right)^{\epsilon} Y_t$$

integrating over i and rearranging we get that aggregate production is

$$Y_t = A_t\,(K_t)^{\alpha}\,(L_t)^{1-\alpha}$$

C.4 Labor supply

- Let households have GHH preferences, then

$$\sum_{t=0}^{\infty} \beta^t \frac{\left(c - \frac{L^{1+\nu}}{1+\nu}\right)^{1-\sigma}}{1-\sigma}$$

  the F.O.C. with respect to labor implies

$$L^{\nu} = w_t$$

  recall the capital-to-labor ratio is

$$\frac{K_t}{L_t} = \frac{\alpha}{1-\alpha}\frac{w_t}{r_t}$$

  plugging in and rearranging

$$L^{1+\nu} = \frac{1-\alpha}{\alpha} K_t r_t$$

- Now, recall from the individual capital demand we had

$$k_{it} = \frac{\alpha}{r_t}\frac{\epsilon - 1}{\epsilon}\left(p_{it}\right)^{1-\epsilon}\left(P_t\right)^{\epsilon} Y_t$$

  integrating over i and using market clearing and the definition of P_t we get

$$K_t = \frac{\alpha}{r_t}\frac{\epsilon - 1}{\epsilon} P_t Y_t$$

  normalizing $P_t = 1$, plugging in for Y_t and rearranging we get

$$K_t r_t = \alpha \frac{\epsilon - 1}{\epsilon} A_t \left(K_t\right)^{\alpha}\left(L_t\right)^{1-\alpha}$$

- Plugging into the household's optimality condition from labor supply we get

$$L_t = \left[(1-\alpha)\frac{\epsilon - 1}{\epsilon} A_t \left(K_t\right)^{\alpha}\right]^{\frac{1}{\alpha+\nu}}$$

the labor supply L_t as a function of the state (z_t, K_t) and of the cutoff z_t^* (note A_t depends on the cutoff)

- So output is

$$\begin{aligned} Y_t &= A_t \left(K_t\right)^{\alpha} \left(L_t\right)^{1-\alpha} \\ &= \left[\left(1-\alpha\right)\frac{\epsilon-1}{\epsilon}\right]^{\frac{1-\alpha}{\alpha+\nu}} \left(A_t\right)^{\frac{1+\nu}{\alpha+\nu}} \left(K_t\right)^{\alpha\left(\frac{1+\nu}{\alpha+\nu}\right)} \end{aligned}$$

C.5 Adoption cutoff

- Operating profits of a firm i are

$$\pi_{it}^{O} = \frac{1}{\epsilon-1}\frac{\mu_t}{z_{it}} y_{it}$$

plugging in for $y_{it} = \left(\frac{P_t}{p_{it}}\right)^{\epsilon} Y_t$ and using the normalization $P_t = 1$ we get

$$\pi_{it}^{O} = \frac{1}{\epsilon-1}\frac{\mu_t}{z_{it}}\left(\frac{1}{p_{it}}\right)^{\epsilon} Y_t$$

plugging in for $p_{it} = \frac{\epsilon}{\epsilon-1}\frac{\mu_t}{z_{it}}$ we get

$$\pi_{it}^{O} = \frac{1}{\epsilon}\left(\frac{\epsilon-1}{\epsilon}\frac{z_{it}}{\mu_t}\right)^{\epsilon-1} Y_t$$

- The productivity cutoff z_t^* satisfies

$$\pi_{it}^{OT} = \pi_{it}^{OM} - f_m\left(1+r_t\right)$$

plugging in for π_{it}^{OT}

$$\frac{1}{\epsilon}\left(\frac{\epsilon-1}{\epsilon}\frac{z_t z_t^* z_T}{\mu_t}\right)^{\epsilon-1} Y_t = \frac{1}{\epsilon}\left(\frac{\epsilon-1}{\epsilon}\frac{z_t z_t^*}{\mu_t}\right)^{\epsilon-1} Y_t - f_m\left(1+r_t\right)$$

rearranging

$$\left(\frac{\epsilon-1}{\epsilon}\frac{z_t z_t^*}{\mu_t}\right)^{\epsilon-1}\left[1-\left(z_T\right)^{\epsilon-1}\right] = \frac{\epsilon f_m\left(1+r_t\right)}{Y_t}$$

plugging in for $Y_t = \left[(1-\alpha)\frac{\epsilon-1}{\epsilon}\right]^{\frac{1-\alpha}{\alpha+\nu}} (A_t)^{\frac{1+\nu}{\alpha+\nu}} (K_t)^{\alpha\left(\frac{1+\nu}{\alpha+\nu}\right)}$ and rearranging:

$$\left(z_t^*\right)^{\epsilon-1} = \frac{\epsilon f_m (1+r_t)}{(A_t)^{\frac{1+\nu}{\alpha+\nu}} (1-\alpha)^{\frac{1-\alpha}{\alpha+\nu}} (K_t)^{\alpha\left(\frac{1+\nu}{\alpha+\nu}\right)} \left[1-(z_T)^{\epsilon-1}\right]} \left(\frac{\epsilon}{\epsilon-1}\right)^{\frac{1-\alpha}{\alpha+\nu}+(\epsilon-1)} \left(\frac{\mu_t}{z_t}\right)^{\epsilon-1}$$

- Note that μ_t is a function of A_t:

$$\begin{aligned}
\mu_t &= \left(\frac{w_t}{1-\alpha}\right)^{1-\alpha} \left(\frac{r_t}{\alpha}\right)^{\alpha} \\
w_t &= \frac{1-\alpha}{\alpha}\frac{K_t}{L_t} r_t \\
r_t &= \alpha \frac{\epsilon-1}{\epsilon} A_t (K_t)^{\alpha-1} (L_t)^{1-\alpha} \\
L_t &= \left[(1-\alpha)\frac{\epsilon-1}{\epsilon} A_t (K_t)^{\alpha}\right]^{\frac{1}{\alpha+\nu}}
\end{aligned}$$

plugging in and rearranging

$$\mu_t = \frac{\epsilon-1}{\epsilon} A_t$$

- Plugging for μ_t and rearranging we get

$$\left(z_t^*\right)^{\epsilon-1} = \frac{\epsilon f_m (1+r_t)}{(1-\alpha)^{\frac{1-\alpha}{\alpha+\nu}} (K_t)^{\alpha\left(\frac{1+\nu}{\alpha+\nu}\right)} \left[1-(z_T)^{\epsilon-1}\right]} \left(\frac{\epsilon}{\epsilon-1}\right)^{\frac{1-\alpha}{\alpha+\nu}} \left(\frac{1}{z_t}\right)^{\epsilon-1} (A_t)^{(\epsilon-1)-\frac{1+\nu}{\alpha+\nu}}$$

- Note that A_t is decreasing in z_t^*. If we assume $(\epsilon-1) > \frac{1+\nu}{\alpha+\nu}$ then the RHS is decreasing in z_t^*. As $z_t^* \to \infty$ we get

$$\infty > \frac{\epsilon f_m (1+r_t)}{(1-\alpha)^{\frac{1-\alpha}{\alpha+\nu}} (K_t)^{\alpha\left(\frac{1+\nu}{\alpha+\nu}\right)} \left[1-(z_T)^{\epsilon-1}\right]} \left(\frac{\epsilon}{\epsilon-1}\right)^{\frac{1-\alpha}{\alpha+\nu}} \left(\frac{1}{z_t}\right)^{\frac{1+\nu}{\alpha+\nu}} \left(z_T \left(\frac{\xi}{1+\xi-\epsilon}\right)^{\frac{1}{\epsilon-1}}\right)^{(\epsilon-1)-\frac{1+\nu}{\alpha+\nu}}$$

so the LHS is higher with high enough cutoff z_t^*. This means that there's always a positive mass of firms adopting the modern technology. To have an interior cutoff we need

$$1 < \frac{\epsilon f_m (1+r_t)}{(1-\alpha)^{\frac{1-\alpha}{\alpha+\nu}} (K_t)^{\alpha\left(\frac{1+\nu}{\alpha+\nu}\right)} \left[1-(z_T)^{\epsilon-1}\right]} \left(\frac{\epsilon}{\epsilon-1}\right)^{\frac{1-\alpha}{\alpha+\nu}} \left(\frac{1}{z_t}\right)^{\frac{1+\nu}{\alpha+\nu}} \left(\left(\frac{\xi}{1+\xi-\epsilon}\right)^{\frac{1}{\epsilon-1}}\right)^{(\epsilon-1)-\frac{1+\nu}{\alpha+\nu}}$$

otherwise we get the corner solution of full adoption $z_t^* = 1$. This could happen with a large enough stock of capital or a large enough productivity shock z_t.

- If the above inequality does not hold then the equation

$$\left(z_t^*\right)^{\epsilon-1} = \frac{\epsilon f_m \left(1+r_t\right)}{\left(1-\alpha\right)^{\frac{1-\alpha}{\alpha+\nu}} \left(K_t\right)^{\alpha\left(\frac{1+\nu}{\alpha+\nu}\right)} \left[1-\left(z_T\right)^{\epsilon-1}\right]} \left(\frac{\epsilon}{\epsilon-1}\right)^{\frac{1-\alpha}{\alpha+\nu}} \left(\frac{1}{z_t}\right)^{\epsilon-1} \left(A\left(z_t^*\right)\right)^{(\epsilon-1)-\frac{1+\nu}{\alpha+\nu}}$$

pins down $z_t^* \in (1, \infty)$. Once we know z_t^* we know A_t and all other variables.